

TICVAI Device Management

Functional Requirements and Dashboard Design Specification

Eight boards covering device registration, configuration, monitoring, maintenance, software, security, integrations, and enterprise operations

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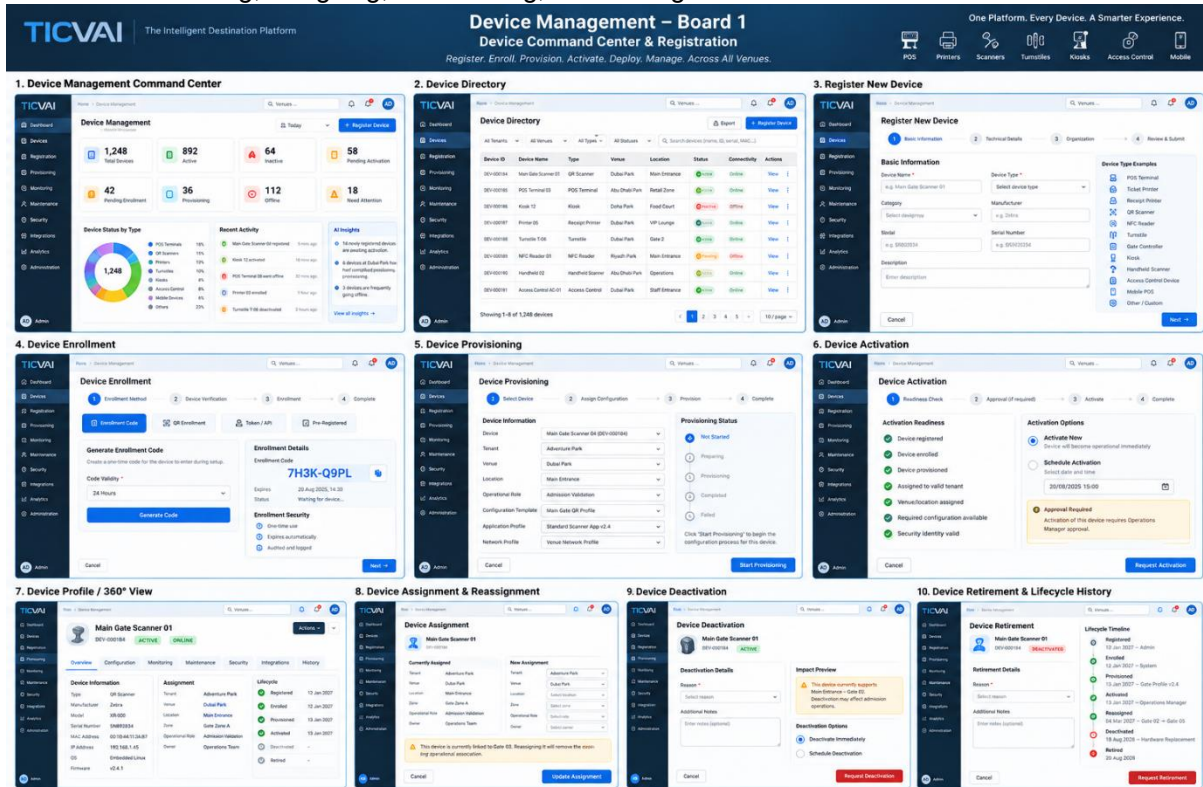
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Board 1 Device Command Center and Registration

Board Objective: Provide the central operational workspace for registering, enrolling, provisioning, activating, assigning, deactivating, and retiring all devices connected to TICVAI.



This board primarily covers **16.1 Device Registration** and the core device-management foundation required by **16.9 Enterprise Features**. The source specifically requires registration, enrollment, provisioning, activation, deactivation and retirement.

Screen 1 Device Management Command Center

Purpose

Provide administrators with a real-time executive and operational overview of all devices across tenants, clients, venues and locations.

Functional Scope

Dashboard shall display:

- Total registered devices
- Active devices
- Inactive devices
- Pending activation
- Pending enrollment
- Provisioning in progress
- Offline/unreachable devices
- Devices requiring attention
- Recently registered devices
- Recently deactivated/retired devices

Device Breakdown

Allow breakdown by:

- Tenant / Client

- Venue
- Location / Zone
- Device type
- Manufacturer/model
- Operational status
- Ownership
- Environment
- Registration method

Quick Actions

Authorized users can:

+ Register Device

Bulk Register

View Inventory

Pending Activations

Import Devices

AI Insight Panel

TICVAI AI may highlight operational observations such as:

“14 newly registered devices are still awaiting activation.”

“6 devices registered for Venue A have not completed provisioning.”

AI recommendations are advisory only and must respect RBAC.

Screen 2 Device Directory

Purpose

Provide the master searchable directory of all devices registered within TICVAI.

Table Fields

Field	Example
Device ID	DEV-000184
Device Name	Main Gate Scanner 01
Device Type	QR Scanner
Tenant	Adventure Park
Venue	Dubai Venue
Location	Main Entrance
Manufacturer	Vendor X
Model	XR-500
Serial Number	SN892834
Status	Active
Connectivity	Online
Assigned To	Gate 01
Last Seen	2 mins ago

Filters

Filter by:

- Tenant
- Venue
- Location
- Device type
- Status
- Manufacturer
- Model
- Ownership
- Registration date
- Activation status

Support keyword search by device ID, name, serial number, MAC/device identifier, etc.

Actions

View | Edit | Assign | Activate | Deactivate | More

Bulk selection shall also be supported.

Screen 3 Register New Device**Purpose**

Provide the main workflow for manually registering a new device.

Registration Form**Basic Information**

- Device name
- Device type
- Device category
- Manufacturer
- Model
- Serial number
- Asset/device ID
- Description

Technical Identity

- MAC address
- IP address
- Hardware identifier
- UUID
- Hostname
- OS/platform where applicable

Not every field should be mandatory. Required fields shall be configurable according to device type.

Organizational Context

Assign:

Tenant → Venue → Location → Zone

Device Types

Configurable device types can include:

- POS terminal

- Ticket printer
- Receipt printer
- QR scanner
- NFC reader
- RFID reader
- Handheld scanner
- Turnstile
- Gate controller
- Kiosk
- Game/Ride reader
- Mobile POS
- Access-control device
- Other/custom device

This prepares the architecture for the integrations explicitly required later in 16.8.

Screen 4 Device Enrollment

Purpose

Manage the secure process through which a physical device establishes itself as an authorized TICVAI device.

Enrollment Methods

Support configurable methods such as:

Enrollment Code

Administrator generates a one-time code entered on the device.

QR Enrollment

TICVAI generates a secure enrollment QR that the device/application scans.

Token and API Enrollment

Used for compatible integrated hardware.

Pre Registered Enrollment

Serial number/device identity is preloaded and automatically matched when the device connects.

Enrollment Status

Lifecycle:

Not Enrolled → Enrollment Requested → Identity Verified → Enrolled

Security

Capture:

- Enrollment initiator
- Device identity
- Date/time
- IP/network
- authentication result
- failed attempts

Enrollment tokens/codes shall expire based on configurable security rules.

Screen 5 Device Provisioning

Purpose

Configure what the newly enrolled device is supposed to do within TICVAI.

Provisioning Information

Assign:

- Tenant
- Venue
- Location
- Operational role
- Device profile
- Configuration template
- Application/software profile
- Network profile where applicable

Example:

Device: Main Gate Scanner 04

Venue: Theme Park Dubai

Location: Main Entrance

Role: Admission Validation

Configuration: Main Gate QR Profile

Provisioning Status

Not Started → **Preparing** → **Provisioning** → **Completed / Failed**

Actions

Start Provisioning

Retry

Cancel

View Provisioning Log

Detailed configuration editing itself will be handled in **Board 2 – Device Inventory & Configuration**, avoiding duplication.

Screen 6 Device Activation

Purpose

Control when a registered and provisioned device becomes operational.

Activation Readiness

Before activation, TICVAI verifies:

- ✓ Registered
- ✓ Enrolled
- ✓ Provisioned
- ✓ Assigned to valid tenant
- ✓ Venue/location assigned
- ✓ Required configuration available
- ✓ Security identity valid

Activation Actions

Activate Now

or

Schedule Activation

Approval

Activation can optionally require approval depending on:

- Device type
- Venue
- tenant policy
- operational sensitivity

Example:

Gate Controller GC-04

Activation requires Operations Manager approval.

Activation Result

ACTIVE

Record:

- Activated by
- Activation timestamp
- Approved by
- Configuration version
- Device identity
- Venue/location

Screen 7 Device Profile and 360 View

Purpose

Provide the single authoritative record for each device.

Header

Main Gate Scanner 01

ACTIVE | ONLINE

Profile Sections

Overview

- Device ID
- Type
- Manufacturer/model
- Serial number
- Current status

Assignment

- Tenant
- Venue
- Location
- Zone
- Operational role

Technical

- Hardware ID
- IP
- MAC
- OS
- firmware
- software version

Lifecycle

- Registered
- Enrolled
- Provisioned
- Activated
- Deactivated
- Retired

Related Tabs

Overview | Configuration | Monitoring | Maintenance | Security | Integrations | History

The profile acts as a navigation hub. Detailed functionality belongs to the appropriate later boards.

Screen 8 Device Assignment and Reassignment

Purpose

Allow administrators to control where and how a device is deployed.

Current Assignment

Display:

Tenant: Adventure Park

Venue: Dubai Park

Location: Main Entrance

Zone: Gate Zone A

Operational Role: Admission Scanner

Reassignment

Allow:

- Change tenant where permitted
- Change venue
- Change location
- Change zone
- Change operational role
- Assign/unassign responsible owner

Validation

Before reassignment, TICVAI checks dependencies.

Example:

This device is currently linked to Gate 03. Reassigning it will remove the existing operational association.

User must confirm governed changes.

All reassignment actions must be audited.

Screen 9 Device Deactivation

Purpose

Temporarily remove a device from operational use without deleting or retiring it.

Deactivation Reasons

Configurable reasons such as:

- Maintenance
- Fault
- Security concern
- Venue closure

- Replacement
- Temporary shutdown
- Configuration issue
- Other

Options

Deactivate Immediately

or

Schedule Deactivation

Impact Preview

Before confirmation show:

Device currently supports Main Entrance – Gate 02.

Deactivation may affect admission operations.

Workflow

Active → Deactivation Requested → Approved → Deactivated

Approval can be enabled based on governance rules.

Reactivation

A deactivated device can later be:

Reactivated

without creating a new device record.

Screen 10 Device Retirement and Lifecycle History

Purpose

Permanently remove a device from active operational service while preserving its complete history.

Retirement Reasons

- End of life
- Replaced
- Damaged
- Lost
- Obsolete
- Returned to vendor
- Contract ended
- Security retirement
- Other

Pre Retirement Check

TICVAI identifies dependencies:

Device is assigned to Gate 05.

Device has an active configuration profile.

Device has unresolved maintenance activity.

Retirement Workflow

Request Retirement → Impact Check → Approval → Revoke Credentials → Unassign → Retired

Retirement **must not delete the device**.

Lifecycle Timeline

Example:

Registered

12 Jan 2027 – Admin

**Enrolled**

12 Jan 2027 – Secure Enrollment

**Provisioned**

13 Jan 2027 – Gate Profile V2.4

**Activated**

13 Jan 2027 – Operations Manager

**Reassigned**

04 Mar 2027 – Gate 02 → Gate 05

**Deactivated**

18 Aug 2028 – Hardware Replacement

**Retired**

20 Aug 2028

Complete audit history remains searchable.

Board 1 Shared Configuration Requirements

The backend shall allow authorized administrators to configure:

- Device types and categories
- Device naming/ID rules
- Required registration fields by device type
- Enrollment methods
- Enrollment token expiry
- Provisioning rules
- Activation prerequisites
- Activation/deactivation approval rules
- Retirement approval rules
- Deactivation and retirement reason codes
- Tenant/venue/location hierarchy
- Device ownership models
- Device operational roles
- Lifecycle statuses
- Bulk registration rules
- API-based registration permissions
- RBAC permissions

Bulk and API Registration

The matrix requires device-management APIs at enterprise level.

Therefore Board 1 should support:

CSV/Excel Import | Bulk Registration | API Registration

with validation, duplicate detection, error reporting and import results.

Roles and Permissions

Typical permissions:

Super Admin — all tenants/devices

Tenant Admin — tenant devices

Venue Admin — venue devices

IT Administrator — technical registration/provisioning

Operations Manager — operational assignment/activation

Maintenance User — view device and maintenance context

Viewer/Auditor — read-only

All critical actions must be governed through TICVAI RBAC/PBAC and audit logging.

Board 1 Acceptance Criteria

Board 1 is accepted when an authorized user can successfully complete the full lifecycle:

Register → Enroll → Provision → Activate → Assign/Reassign → Deactivate/Reactivate → Retire

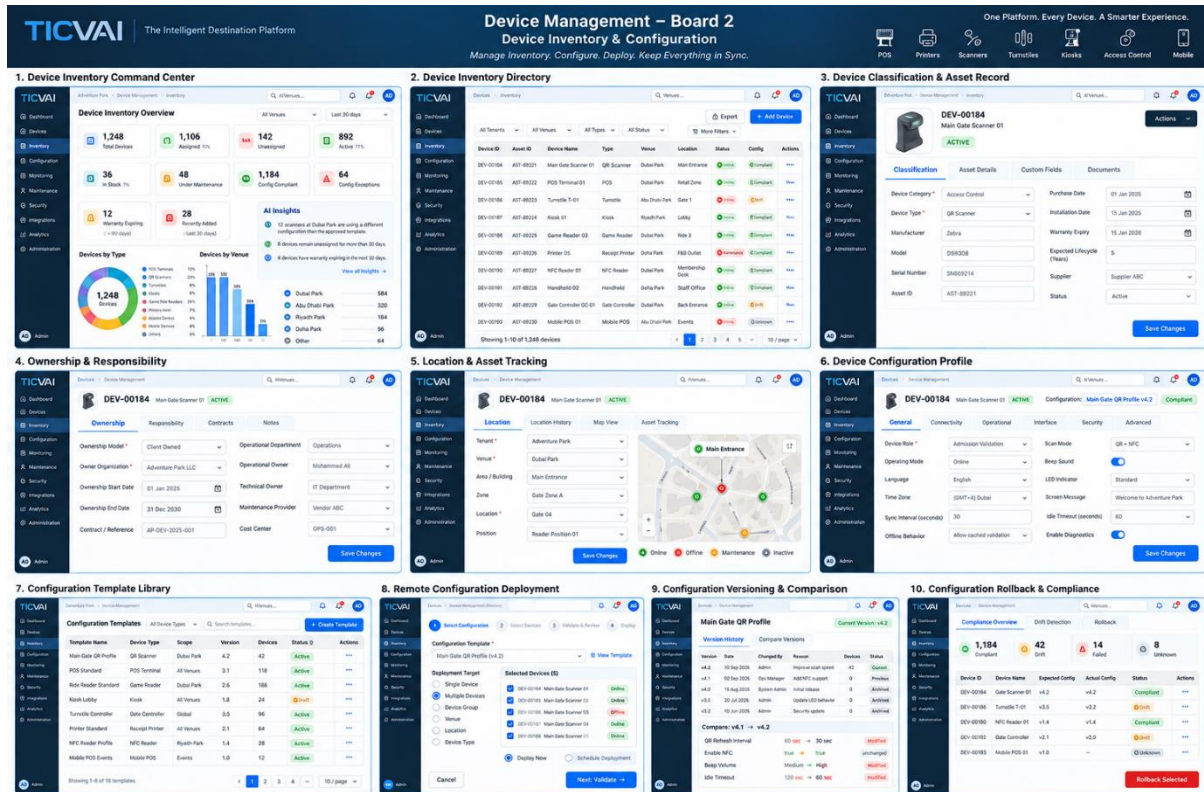
The system must preserve a single device identity throughout that lifecycle, enforce tenant and venue isolation, prevent unauthorized actions, detect duplicate registrations, validate dependencies before lifecycle changes, support bulk/API registration, and maintain a complete audit trail.

Board 1 = 10 screens.

This gives us a strong foundation. Importantly, I kept **monitoring, detailed configuration, firmware, maintenance, security and integration management out of Board 1** except where their status/context must be visible. Those belong to Boards 2–8, which will prevent the duplication problem we encountered in some of the earlier TICVAI modules.

Board 2 Device Inventory and Configuration

Board Objective: Provide TICVAI administrators with a centralized workspace to manage the **inventory, classification, ownership, assignment, physical location, technical configuration, reusable configuration templates, configuration versions, deployment status, and rollback** of every registered device.



Board 2 primarily covers **16.2 Device Inventory** and **16.3 Device Configuration**. The matrix requires device inventories, classification, ownership, assignment, location and asset tracking, followed by configuration management, remote configuration, reusable templates, versioning and rollback.

The key principle is:

Board 1 manages the device lifecycle. Board 2 manages what the device is, where it belongs, and how it is configured.

Screen 1 Device Inventory Command Center

Purpose

Provide a centralized operational view of the complete hardware/device inventory across all TICVAI tenants, venues and locations.

KPI Cards

Display:

- Total Devices
- Assigned Devices
- Unassigned Devices
- Active Devices
- In Stock
- Under Maintenance
- Configuration Compliant
- Configuration Exceptions

- Warranty Expiring
- Recently Added

Inventory Breakdown

Interactive charts/cards by:

Device Type | Tenant | Venue | Location | Manufacturer | Ownership | Status

Example:

Device Type	Total	Assigned	Available	Attention
POS	146	139	7	3
QR Scanner	284	270	14	8
Turnstile	96	96	0	4
Kiosk	42	39	3	2
Game/Ride Reader	318	304	14	11

Quick Actions

+ Add Inventory Device

Bulk Import

Assign Devices

Configuration Templates

Export Inventory

AI Insights

Examples:

“12 scanners at Venue A are using a configuration different from the approved venue template.”

“8 devices remain unassigned more than 30 days after registration.”

AI remains advisory.

Screen 2 Device Inventory Directory

Purpose

Provide the detailed master inventory of all physical devices.

Table Fields

Field	Example
Device ID	DEV-00184
Asset ID	AST-89221
Device Name	Main Gate Scanner 01
Category	Access
Type	QR Scanner
Manufacturer	Vendor X
Model	QR-X5
Serial Number	SN889214
Tenant	Adventure Park

Field	Example
Venue	Dubai Park
Location	Main Entrance
Ownership	Client Owned
Status	Active
Config	Compliant

Search

Search by:

- Device ID
- Asset ID
- Serial number
- Device name
- MAC
- Manufacturer/model

Advanced Filters

Tenant
Venue
Location
Zone
Device type
Category
Manufacturer
Ownership
Assignment status
Lifecycle status
Configuration status

Bulk Actions

Users can select multiple devices and:

Assign | Reassign | Apply Template | Update Classification | Export

Permissions must control bulk operations.

Screen 3 Device Classification and Asset Record

Purpose

Define the business and technical classification of each device.

Device Classification

Configure:

Device Category

Examples:

- POS
- Access Control
- Ticketing
- Kiosk
- Payment
- Game/Ride

- Printing
- Mobile
- Network
- Other

Device Type

Example:

Access Control → QR Scanner

Asset Information

Capture:

- Asset ID
- Manufacturer
- Model
- Serial number
- Purchase date
- Installation date
- Supplier
- Warranty start
- Warranty expiry
- Expected lifecycle
- Replacement date
- Internal asset reference

Custom Fields

TICVAI shall allow administrators to define additional metadata by device category.

For example:

Turnstile

- Lane width
- Direction
- Controller type

Game Reader

- Reader model
- Screen type
- LED capability
- NFC/RFID capability

This avoids hardcoding every hardware type into the platform.

Screen 4 Ownership and Responsibility

Purpose

Clearly define who owns and who is operationally responsible for each device.

Ownership Models

Configurable options:

- TICVAI Owned
- Client Owned
- Venue Owned

- Leased
- Rental
- Vendor Managed
- Third-Party Owned

Ownership Record

Capture:

Owner Organization

Tenant

Contract/Reference

Ownership Start Date

Ownership End Date

Responsible Department

Responsible User/Team

Operational Responsibility

Separate ownership from operational responsibility.

Example:

Owner: Adventure Park LLC

Operational Owner: Venue Operations

Technical Owner: IT Department

Maintenance Provider: Vendor ABC

This distinction is important for future maintenance and SLA workflows.

Screen 5 Location and Asset Tracking

Purpose

Track exactly where each device is deployed.

Location Hierarchy

Support:

Tenant → Venue → Building/Area → Zone → Location → Position

Example:

Adventure Park

→ Dubai Venue

→ Main Entrance

→ Gate Zone A

→ Gate 04

→ Reader Position 01

Device Movement

Every movement must create history:

Previous Location → New Location → Changed By → Date/Time → Reason

Map and Venue View

Where venue mapping is available, devices may be displayed graphically.

Example icons:



Online



Offline



Attention



Inactive

Location History

Users can view the full historical movement of the device.

This fulfills the matrix requirement to maintain device location and asset-tracking records.

Screen 6 Device Configuration Profile

Purpose

Provide the central configuration workspace for an individual device.

Configuration Header

Example:

Main Gate Scanner 01

Configuration: Main Gate QR Profile

Version: 4.2

Status: COMPLIANT

Configuration Sections

Depending on device capability:

General

- Device role
- Operating mode
- Language
- Time zone

Connectivity

- Network profile
- Server endpoint
- Port
- Communication protocol

Operational

- Operating parameters
- Local behavior
- Offline behavior
- Sync interval

Interface

- Display behavior
- Sound
- LED behavior

Security

- Security profile reference
- Authentication method

Important Architecture Rule

TICVAI must support **dynamic configuration schemas by device type**.

A POS terminal should not show the same configuration fields as a turnstile controller or game reader.

Screen 7 Configuration Template Library

Purpose

Allow administrators to create reusable configurations instead of manually configuring every device.

Template Directory

Template	Device Type	Venue	Version	Devices	Status
Main Gate QR	QR Scanner	Dubai Park	4.2	42	Active
POS Standard	POS	All Venues	3.1	118	Active
Ride Reader Standard	Game Reader	Dubai Park	2.6	186	Active

Template Scope

Templates may apply at:

Global → Tenant → Venue → Location → Device Group

Create Template

Administrator defines:

- Template name
- Device type
- Supported models
- Configuration parameters
- Scope
- Effective date
- Approval requirement

Template Inheritance

Example:

Global QR Template

↓

Adventure Park QR Template

↓

Dubai Venue QR Template

↓

Main Entrance Override

This allows TICVAI to operate at enterprise scale without creating hundreds of duplicate configurations.

Screen 8 Remote Configuration Deployment

Purpose

Allow authorized administrators to remotely deploy approved configurations to one or many devices.

Remote configuration is explicitly required by the matrix.

Deployment Workflow

Select Configuration

↓

Select Target Devices

↓

Validate Compatibility

↓

Impact Preview

↓

Approval if Required

↓

Deploy

Target Selection

Deploy to:

- Single device
- Selected devices
- Device group
- Venue
- Location
- Device type

Pre Deployment Validation

TICVAI checks:

- ✓ Device compatibility
- ✓ Device online status
- ✓ Supported configuration version
- ✓ Security authorization
- ✓ Existing configuration
- ✓ Dependencies

Deployment Options

Deploy Now

or

Schedule Deployment

Deployment Status

Show:

Queued | Sending | Applying | Successful | Failed | Offline/Pending

Offline devices can receive the approved configuration once connectivity is restored, subject to policy.

Screen 9 Configuration Versioning and Comparison

Purpose

Maintain a complete historical record of device configurations.

Version History

Version	Date	Changed By	Status	Devices
V4.2	10 Sep 2026	Admin	Current	42
V4.1	02 Sep 2026	Operations	Previous	0
V4.0	15 Aug 2026	System Admin	Archived	0

Compare Versions

Allow side-by-side comparison:

V4.1 ↔ V4.2

Highlight:

Added Parameters

Removed Parameters

Changed Values

Example:

QR Refresh Interval

V4.1: 60 sec

V4.2: 30 sec

Change Information

Every version records:

- Created by
- Date/time
- Reason
- Approval
- Effective date
- Change summary

Audit

Published configuration versions must be immutable.

Any change creates a **new version**.

Screen 10 Configuration Rollback and Compliance

Purpose

Allow administrators to identify configuration drift and safely restore an approved configuration.

Compliance Dashboard

Show devices as:

☒ **Compliant**

Device matches approved configuration.

☒ **Configuration Drift**

Device configuration differs from approved template.

☒ **Deployment Failed**

Configuration could not be applied.

☐ **Unknown**

Configuration status cannot currently be verified.

Drift Detection

Example:

Main Gate Scanner 04

Expected: Main Gate QR Profile V4.2

Actual: V4.1

Differences:

Refresh Interval: 60 sec → Expected 30 sec

Actions

View Difference

Apply Current Template

Rollback

Create Exception

Rollback

Administrator selects a previously approved configuration:

Current V4.2 → Rollback to V4.1

Before execution, show:

- Target devices

- affected locations
- configuration differences
- operational impact
- compatibility
- approval requirement

Emergency Rollback

Authorized users may execute emergency rollback where configured.

The system must record the reason and full audit trail.

Board 2 Shared Configuration Requirements

TICVAI administrators shall be able to configure:

- Device categories
- Device types/subtypes
- Manufacturer/model catalogue
- Asset numbering rules
- Ownership types
- Responsibility types
- Venue/location hierarchy
- Custom device metadata
- Configuration schemas by device type
- Configuration templates
- Template hierarchy/inheritance
- Device grouping rules
- Mandatory configuration parameters
- Configuration validation rules
- Deployment permissions
- Deployment approval rules
- Scheduled deployments
- Configuration version rules
- Drift detection rules
- Rollback permissions
- Emergency rollback permissions

Configuration Hierarchy

I recommend Softlabs build the configuration engine around the following hierarchy:

TICVAI Global Default

↓

Tenant Configuration

↓

Venue Configuration

↓

Location / Device Group

↓

Individual Device Override

This is extremely important.

For example, if a client has **500 identical game readers**, the administrator should not configure 500 devices individually.

They create:

Game Reader Standard Profile V3 2

and apply it to all 500 devices.

If 20 readers at one attraction require different LED or operational behavior, they receive a location-level override.

This will make the Device Manager scalable.

RBAC and Governance

Permissions should distinguish between:

View Inventory

Edit Asset Information

Move/Reassign Device

Create Configuration

Edit Configuration

Publish Configuration

Deploy Configuration

Rollback Configuration

Emergency Rollback

High-impact changes can require approval.

Example:

IT Admin creates configuration → Operations Manager approves → System deploys.

All actions must maintain:

Who | What | Before | After | When | Reason | Approval

Board 2 Acceptance Criteria

Board 2 is accepted when TICVAI can:

Inventory → Classify → Assign Ownership → Locate → Configure → Template → Deploy → Version → Compare → Detect Drift → Rollback

for individual devices and device groups.

The platform must support multi-tenant and multi-venue isolation, reusable configuration templates, hierarchical configuration inheritance, remote configuration, scheduled deployment, configuration versioning, compatibility validation, rollback and complete audit history.

Board Boundary

To avoid duplication:

Board 1: registration and lifecycle

Board 2: inventory and configuration

Board 3: health, connectivity, status, battery, performance and alerts

Board 4: maintenance

Board 5: firmware/software

Board 6: security

Board 7: hardware integrations

Board 8: enterprise operations, analytics, APIs and SLA

Board 2 = 10 screens.

Board 3 Device Monitoring Health and Alert Management

Board Objective: Provide TICVAI with a centralized, real-time operational monitoring layer for all connected devices, allowing authorized teams to understand **device health, connectivity, operational status, battery condition, performance, failures, and alerts** across tenants, venues and locations.



Board 3 primarily covers **16.4 Device Monitoring**, which requires health monitoring, connectivity monitoring, status monitoring, battery monitoring where applicable, performance monitoring and device alerts.

The key principle is:

Board 3 answers three questions continuously: Is the device reachable? Is it healthy? Does someone need to act?

Screen 1 Device Monitoring Command Center

Purpose

Provide the main real-time operational dashboard for monitoring the entire TICVAI device estate.

Global KPI Cards

Display:

- Total Monitored Devices
- Online
- Offline
- Degraded
- Critical
- Maintenance Mode
- Unreachable
- Active Alerts
- Critical Alerts
- SLA At Risk

Example:

Status	Devices
 Online	1,184
 Offline	34
 Degraded	18
 Maintenance	12
 Critical	7

Health Overview

Provide overall health score:

Device Estate Health: 96.4%

Break down by:

Tenant | Venue | Location | Device Type | Manufacturer | Model

Operational Map

Where venue maps are available, show devices geographically or on the venue layout.

Green = Healthy

Yellow = Warning

Red = Critical/Offline

Grey = Inactive/Maintenance

Critical Attention Panel

Example:

-  Gate Controller GC-04 offline for 14 minutes.
-  8 handheld scanners below 20% battery.
-  Ride Reader RR-018 reporting repeated communication failures.

Quick Actions

View Critical Devices | Alert Center | Device Map | Monitoring Rules | Export Status

Screen 2 Live Device Status Monitor

Purpose

Provide an operational table showing the current state of every monitored device.

Table Fields

Field	Example
Device	Main Gate Scanner 01
Type	QR Scanner
Venue	Dubai Park
Location	Gate 01
Health	Healthy
Connectivity	Online
Operational Status	Running
Battery	82%

Field	Example
Last Seen	8 sec ago
Alerts	0
Response	42 ms

Status Indicators

Health

-  Healthy
-  Warning
-  Degraded
-  Critical
-  Unknown

Connectivity

-  Online
-  Offline
-  Intermittent
-  Unknown

Filters

Allow filtering by:

- Tenant
- Venue
- Location
- Device type
- Health
- Connectivity
- operational status
- battery level
- alert severity
- manufacturer/model

Refresh

Support:

Real Time and Auto Refresh and Manual Refresh

Users can open the Device 360° Monitoring View directly from this screen.

Screen 3 Device Health 360 View

Purpose

Provide detailed health information for a single device.

Header

Main Gate Scanner 01

ONLINE HEALTHY

Health Score: 98%

Health Components

Depending on device capability:

- Connectivity

- CPU
- Memory
- Storage
- Temperature
- Battery
- application/service status
- peripheral status
- controller status
- sensor status
- communication quality
- error frequency

Example:

Component	Status
Connectivity	 Healthy
CPU	 34%
Memory	 51%
Storage	 62%
Temperature	 41°C
Application	 Running
Scanner	 Operational

Timeline

Show health over:

1H | 6H | 24H | 7D | 30D

Device Context

Display:

Tenant → Venue → Location → Device

This screen is monitoring-focused and links back to Board 1's master device profile rather than duplicating it.

Screen 4 Connectivity Monitoring

Purpose

Monitor whether devices can reliably communicate with TICVAI and their required services.

Connectivity Metrics

Where technically available:

- Online/offline
- Last heartbeat
- Last successful communication
- Latency
- packet loss
- connection stability
- reconnect count

- network type
- IP
- communication endpoint

Connectivity Timeline

Example:

08:00–10:42 — Online

10:42–10:47 — Offline

10:47–11:32 — Online

Flapping Detection

TICVAI should identify devices repeatedly moving between online and offline states.

Example:

⚠ Device GC-018 disconnected 17 times during the last 60 minutes.

Dependency Test

Where supported:

Device → Local Controller → Network → TICVAI Platform

This helps operations distinguish between:

Device problem

versus

Local network/controller problem.

Actions

Ping and Test Connection

View Communication Logs

Open Incident

Place in Maintenance Mode

Screen 5 Battery and Power Monitoring

Purpose

Monitor battery and power condition for applicable handheld, mobile, wireless and portable devices.

The matrix explicitly specifies battery monitoring **where applicable**, so this screen must dynamically appear only for supported devices.

Dashboard

Show:

- Battery %
- Charging status
- Power source
- Estimated remaining runtime
- battery health
- charging history
- low battery events

Battery Thresholds

Example:

● **Normal:** > 40%

● **Low:** 20–40%

● **Critical:** < 20%

Thresholds must be configurable.

Fleet Battery View

Example:

186 handheld devices

142 Ready

26 Charging

13 Low

5 Critical

Predictive Warning

Where sufficient telemetry exists, AI can identify patterns such as:

“Handheld HH-042 is unlikely to complete today's operating shift based on current discharge rate.”

This is advisory and should show the supporting telemetry.

Screen 6 Device Performance Monitoring

Purpose

Monitor technical and operational performance indicators for supported devices.

Technical Metrics

Depending on hardware:

- CPU utilization
- memory utilization
- storage utilization
- response time
- processing latency
- network latency
- uptime
- restart count
- application/service performance
- error rate

Operational Metrics

Device-specific metrics may include:

QR Scanner

- Scans/minute
- successful reads
- failed reads

Turnstile

- transactions/entries
- opening response time
- controller errors

POS

- transactions
- peripheral availability

Game and Ride Reader

- validations
- reader response time

- communication failures

Trend Graphs

Display:

Current | 1H | 24H | 7D | 30D

Important Design Principle

Performance metrics must be **device-type configurable**.

We should not hardcode the same monitoring metrics for every hardware type.

Screen 7 Alert Center

Purpose

Provide one centralized location for all device-generated alerts.

Alert Table

Severity	Device	Alert	Venue	Since	Status
Critical	GC-04	Device Offline	Dubai Park	14 min	Open
High	RR-018	Communication Failure	Ride Zone	8 min	Open
Medium	HH-042	Battery Low	Entrance	12 min	Acknowledged

Severity Levels

Configurable:

Critical

High

Medium

Low

Information

Alert Lifecycle

New → Acknowledged → Assigned → Investigating → Resolved → Closed

Actions

Acknowledge

Assign

Create Incident

Mute and Suppress

Resolve

View Device

Alert Ownership

Alerts can be routed to:

- IT
- Operations
- Maintenance
- Venue team
- External vendor
- Specific user/team

Screen 8 Monitoring and Alert Rule Studio

Purpose

Allow authorized administrators to define exactly what TICVAI should monitor and when an alert should be generated.

Example Rule

Rule Name: Gate Controller Offline

Device Type: Gate Controller

Condition:

Connectivity = Offline

Duration:

> 60 seconds

Severity:

Critical

Applies To:

All Production Venues

Additional Rules

Examples:

Battery 20 for 5 minutes

→ High Alert

CPU 90 for 10 minutes

→ Warning

No heartbeat for 60 seconds

→ Critical

Scanner error rate 10

→ High

Temperature Device Threshold

→ Critical

Rule Scope

Rules can apply at:

Global → Tenant → Venue → Device Type → Device Group → Individual Device

Rule Actions

When triggered:

- Create alert
- Notify users
- create incident
- trigger webhook
- escalate
- execute approved automated action where permitted

Screen 9 Alert Escalation and Notification Management

Purpose

Determine who receives an alert, through which channel, and how unresolved alerts escalate.

Escalation Example**Critical Gate Controller Offline****0 minutes**

→ Venue Operations

5 minutes unresolved

→ IT Support

15 minutes unresolved

→ Operations Manager

30 minutes unresolved

→ Management / External Support

Notification Channels

Architecture should support configurable channels such as:

- In-platform notification
- Email
- SMS provider
- Push notification
- Webhook
- External service desk

Escalation Rules

Configure based on:

- Severity
- Tenant
- Venue
- device type
- alert type
- operating hours
- duration
- acknowledgement status

Suppression

Support:

Maintenance Window**Temporary Mute****Planned Downtime**

This is critical so scheduled maintenance does not generate hundreds of false alerts.

Screen 10 Monitoring History AI Insights and Device Availability**Purpose**

Provide historical monitoring analysis and help operations identify recurring problems rather than only responding to individual alerts.

Availability Metrics

Display:

- Device uptime %
- downtime
- number of outages

- average outage duration
- alert count
- critical incidents
- connectivity stability
- health trend

Example

Gate Controller GC 04

Availability: 99.72%

Downtime: 2h 04m

Outages: 7

Critical Alerts: 4

Trend Analysis

Compare:

Today vs Yesterday

This Week vs Previous Week

Venue vs Venue

Device vs Device Group

AI Insights

TICVAI AI can identify patterns such as:

“Gate Controllers GC-04, GC-06 and GC-08 show repeated connectivity loss between 18:00 and 20:00.”

“QR Scanner QS-021 has a significantly higher read-error rate than other scanners of the same model.”

“Five handheld devices show abnormal battery degradation.”

Recommended Actions

AI may suggest:

Inspect Network

Check Configuration

Schedule Maintenance

Compare Similar Devices

Create Incident

AI recommendations should never automatically make critical operational changes unless explicitly governed by configured automation policies.

Board 3 Shared Configuration Requirements

Authorized administrators shall configure:

- Monitoring enable/disable by device type
- Heartbeat frequency
- Connectivity timeout
- Health parameters
- health score calculation
- battery thresholds
- performance thresholds
- device-specific telemetry
- monitoring intervals
- alert types

- severity levels
- alert conditions
- alert duration thresholds
- alert routing
- escalation rules
- notification channels
- maintenance windows
- alert suppression
- acknowledgement rules
- auto-resolution rules
- retention period for monitoring data
- tenant/venue-specific monitoring policies

Device Health Score

I recommend TICVAI introduce a standardized **Device Health Score from 0–100**.

For example:

Connectivity — 30%

Errors — 25%

Performance — 20%

Hardware/Power — 15%

Application/Service Status — 10%

But these weights should **not be fixed globally**.

For a turnstile controller, connectivity may be extremely important. For a handheld scanner, battery condition may carry more weight.

Therefore:

Health Score Model = Configurable by Device Type

This gives management one simple number while allowing the technical team to drill down into the underlying telemetry.

Monitoring Architecture Requirement

This is an important technical requirement for Softlabs.

TICVAI should not assume every device provides the same telemetry.

The architecture should support:

Device and Reader and Controller

↓

Device Adapter and Integration

↓

Telemetry Ingestion

↓

Normalized Device Monitoring Model

↓

Rules and Alert Engine

↓

TICVAI Monitoring Dashboard

For example, a Chinese game reader may report:

reader_status = 1

while another vendor reports:

health = online

TICVAI should normalize both into:

Operational Status = Online

This will make the Device Manager vendor-independent.

Offline and Heartbeat Logic

Each connected device should maintain a configurable heartbeat where technically supported.

Example:

Heartbeat every: 30 seconds

If one heartbeat is missed:

Potential Communication Issue

If heartbeat is absent for 60 seconds:

Offline

If absent for 5 minutes:

Critical Alert

These values must be configurable by device/device type.

When the device reconnects:

Offline → Reconnected

and the outage duration is recorded automatically.

RBAC and Governance

Permissions should distinguish between:

View Monitoring

View Technical Telemetry

Acknowledge Alerts

Assign Alerts

Resolve Alerts

Configure Monitoring Rules

Configure Alert Thresholds

Mute Alerts

Configure Escalation

Execute Device Actions

All rule changes must maintain an audit trail.

Board 3 Acceptance Criteria

Board 3 is accepted when TICVAI can successfully:

Monitor → Detect → Alert → Acknowledge → Assign → Escalate → Resolve → Analyze

device health across supported connected hardware.

The platform must be capable of monitoring health, connectivity, operational status, battery where applicable, and device performance, and generating configurable alerts as required by the matrix.

It must also support multi-tenant/venue isolation, device-specific telemetry, configurable thresholds, real-time status, historical trends, maintenance suppression, escalation and complete audit history.

Board Boundary

Board 1: Device registration & lifecycle

Board 2: Inventory & configuration

Board 3: Monitoring, health & alerts

Board 4: Maintenance & lifecycle servicing

Board 5: Firmware & software management

Board 6: Security & governance

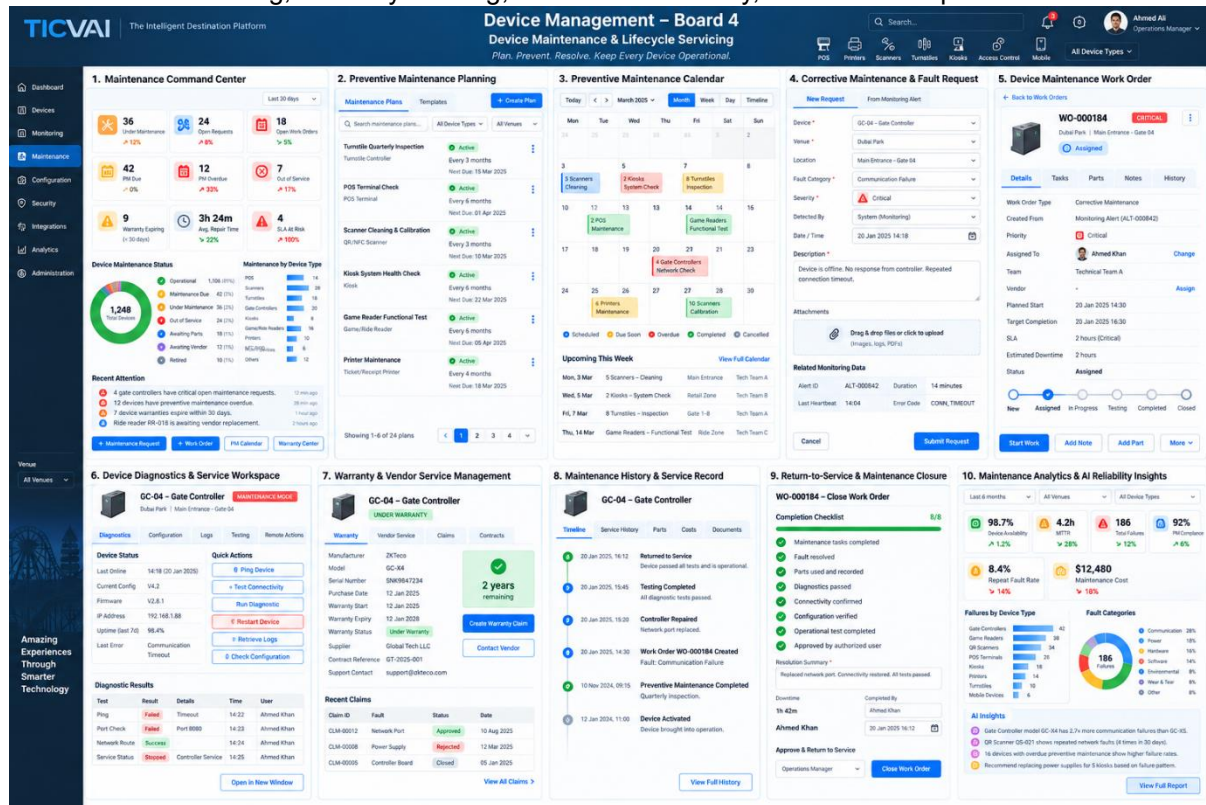
Board 7: Device integrations

Board 8: Enterprise operations, APIs, analytics & SLA

Board 3 = 10 screens.

Board 4 Device Maintenance and Lifecycle Servicing

Board Objective: Provide TICVAI with a centralized operational workspace to manage the maintenance lifecycle of connected devices—from preventive maintenance and fault reporting through work orders, servicing, warranty tracking, maintenance history, and return to operation.



Board 4 primarily covers **16.5 Device Maintenance**: preventive maintenance, corrective maintenance, maintenance work orders, maintenance history, warranty tracking, and device lifecycle tracking.

The key principle is:

Board 3 detects that something is wrong. Board 4 manages what happens next until the device is operational again.

This board should also integrate with the broader **Module 17 – Maintenance & Safety Management**, without duplicating that module. Device Management maintains the **device-specific maintenance view**, while Module 17 provides the enterprise-wide maintenance operation.

Screen 1 Device Maintenance Command Center

Purpose

Provide IT, operations, and maintenance teams with a centralized view of maintenance conditions across the connected device estate.

KPI Cards

Display:

- Devices Under Maintenance
- Open Maintenance Requests
- Open Work Orders
- Preventive Maintenance Due
- Preventive Maintenance Overdue
- Critical Faults
- Devices Out of Service

- Warranty Expiring
- Average Repair Time
- Maintenance SLA at Risk

Maintenance Status Breakdown

Devices categorized as:

- ☒ Operational
- ☐ Maintenance Due
- ☒ Under Maintenance
- ☒ Out of Service
- ☐ Awaiting Parts
- ☒ Awaiting Vendor
- ☒ Retired

Maintenance by Device Type

Break down by:

POS | Scanner | Turnstile | Gate Controller | Kiosk | Game/Ride Reader | Printer | NFC/RFID | Mobile

Attention Panel

Examples:

- ☒ 4 gate controllers have critical open maintenance requests.
- ☒ 12 devices have preventive maintenance overdue.
- ☐ 7 device warranties expire within 30 days.

Quick Actions

+ Maintenance Request

+ Work Order

PM Calendar

Warranty Center

Maintenance History

Screen 2 Preventive Maintenance Planning

Purpose

Define recurring preventive maintenance requirements for devices before failures occur.

Maintenance Plan

Example:

Plan: Turnstile Quarterly Inspection

Applies To: Turnstile Controllers

Frequency: Every 3 Months

Scope: All Production Venues

Responsible Team: Technical Maintenance

Scheduling Methods

Support:

- Daily
- Weekly
- Monthly
- Quarterly

- Annually
- Fixed date
- Usage-based
- Runtime-based
- Manufacturer recommendation
- Custom interval

Where telemetry supports it, maintenance could also be triggered by usage.

Example:

Inspect scanner after every **500,000 scans**.

Maintenance Tasks

A preventive plan may contain:

- ☒ Inspect reader
- ☒ Test connectivity
- ☒ Clean scanner
- ☒ Check power supply
- ☒ Test gate I/O
- ☒ Check cables
- ☒ Run diagnostic test
- ☒ Record result

Assignment

Assign plan to:

Device | Device Group | Device Type | Venue | Location

Screen 3 Preventive Maintenance Calendar

Purpose

Provide a visual planning calendar for upcoming device maintenance.

Calendar Views

Day | Week | Month | Timeline

Maintenance Cards

Example:

15 Sep Main Entrance

8 QR Scanners

Quarterly Inspection

Assigned: Technical Team A

Status

- ☒ Completed
- ☐ Scheduled
- ☐ Due Soon
- ☒ Overdue
- ☒ Cancelled

Filters

Filter by:

- Tenant

- Venue
- location
- device type
- technician/team
- maintenance type
- priority
- status

Rescheduling

Authorized users can:

Reschedule

Reassign

Bulk Reschedule

Changes must be audited.

Screen 4 Corrective Maintenance and Fault Request

Purpose

Create and manage maintenance requests when a device develops a fault.

Create Request

Capture:

Device

Venue and Location

Fault Category

Severity

Description

Detected By

Detection Source

Date and Time

Attachments

Detection Source

A fault can originate from:

Manual Report

Board 3 Monitoring Alert

System Diagnostic

Customer and Venue Report

Technician Inspection

API and External System

Example

Device: Gate Controller GC-04

Fault: Communication Failure

Severity: Critical

Source: Device Monitoring

Duration: 14 minutes

Related Monitoring Data

If generated from Board 3, automatically attach:

- Alert ID

- Health status
- telemetry snapshot
- connectivity history
- error codes
- last heartbeat

This avoids the maintenance technician having to search elsewhere for the technical context.

Screen 5 Device Maintenance Work Order

Purpose

Convert a maintenance requirement into an actionable work order.

Work Order Header

WO-000184

Gate Controller GC 04

CRITICAL

Work Order Information

- Device
- Fault
- Tenant
- Venue
- Location
- Priority
- maintenance type
- assigned technician
- assigned team
- external vendor
- planned start
- target completion
- SLA
- estimated downtime

Task Checklist

Example:

- ☐ Inspect network connection
- ☐ Test controller power
- ☐ Verify TCP/IP connectivity
- ☐ Run diagnostic
- ☐ Inspect I/O connection
- ☐ Restart controller
- ☐ Confirm ticket validation
- ☐ Confirm gate opening

Work Order Lifecycle

New

→ **Assigned**

→ **Accepted**

- In Progress
- Awaiting Parts
- Testing
- Completed
- Closed

Technician Actions

Start Work

Pause

Add Note

Add Part

Upload Photo

Record Test

Complete

Screen 6 Device Diagnostics and Service Workspace

Purpose

Give technicians a device-specific servicing workspace while troubleshooting.

Device Snapshot

Display:

Device: GC-04

Status: Maintenance Mode

Last Online: 14:18

Current Configuration: V4.2

Firmware: V2.8.1

Last Error: Communication Timeout

Diagnostic Actions

Depending on device capability:

Ping Device

Test Connectivity

Run Diagnostic

Restart Device

Test Endpoint

Retrieve Logs

Check Configuration

Check Firmware

Important Governance

Potentially disruptive actions must require permission.

Example:

⚠ Restarting this controller may temporarily affect Gate 04.

Diagnostic Results

Record:

Test | Result | Timestamp | Technician

These results become part of the maintenance history.

Screen 7 Warranty and Vendor Service Management

Purpose

Determine whether a faulty device is covered by warranty or external service support before maintenance cost is incurred.

Warranty Record

Display:

Manufacturer

Supplier

Purchase Date

Warranty Start

Warranty Expiry

Warranty Status

Contract Reference

Support Contact

Status

☒ **Under Warranty**

☐ **Expiring Soon**

☒ **Expired**

Warranty Claim

Allow creation of:

Warranty Claim

with:

- Device
- serial number
- fault
- work order
- supporting files
- vendor
- submission date
- claim reference
- claim status

Claim Lifecycle

Draft → Submitted → Vendor Reviewing → Approved/Rejected → Repair/Replacement → Closed

External Vendor Service

Track:

- Vendor
- pickup date
- expected return
- service status
- vendor notes
- replacement device
- cost where applicable

Screen 8 Maintenance History and Device Service Record

Purpose

Maintain the complete maintenance history of each device.

The matrix explicitly requires maintenance history and lifecycle tracking.

Timeline Example

12 Jan 2027

Device Activated

↓

10 Apr 2027

Preventive Maintenance Completed

↓

18 Aug 2027

Connectivity Fault Detected

↓

18 Aug 2027

WO-00184 Created

↓

18 Aug 2027

Controller Repaired

↓

18 Aug 2027

Testing Passed

↓

18 Aug 2027

Returned to Service

History Table

Date	Type	Issue	Action	Technician	Downtime
18 Aug	Corrective	Network	Port replaced	Ahmed	42m
10 Apr	Preventive	Inspection	Completed	John	15m

Cost History

Where financial integration exists:

- Parts cost
- labor cost
- vendor cost
- total maintenance cost

Screen 9 Return to Service and Maintenance Closure

Purpose

Ensure that a repaired device cannot simply be marked "fixed" without operational validation.

Completion Checklist

Before return to service:

- ✓ Maintenance tasks completed
- ✓ Fault resolved

- ✓ Required parts recorded
- ✓ Diagnostics passed
- ✓ Connectivity confirmed
- ✓ Configuration verified
- ✓ Firmware verified where relevant
- ✓ Operational test completed

Device Specific Testing

For example, a repaired turnstile controller may require:

TCP and IP Test

- ✓ Connected

Ticket Validation

- ✓ Successful

Gate Open Signal

- ✓ Successful

Gate Feedback

- ✓ Received

Closure

Technician:

Complete Maintenance

Operations/authorized user:

Approve Return to Service

Lifecycle

Under Maintenance → Testing → Ready for Service → Operational

If testing fails:

Testing → Maintenance Reopened

Closure Information

Record:

- Root cause
- corrective action
- technician
- completion time
- downtime
- parts used
- final test
- approver

Screen 10 Maintenance Analytics and AI Reliability Insights

Purpose

Provide management with device reliability and maintenance performance intelligence.

KPIs

- Total Failures
- MTBF — Mean Time Between Failures
- MTTR — Mean Time to Repair

- Device Availability
- Preventive Maintenance Compliance
- Repeat Fault Rate
- Maintenance Cost
- Warranty Recovery
- SLA Compliance

Analytics

Analyze by:

Tenant

Venue

Device Type

Manufacturer

Model

Device

Fault Category

Vendor

Reliability Comparison

Example:

Model	Devices	Failure Rate	MTTR	Availability
Reader X5	120	1.8%	22m	99.8%
Reader X4	84	5.4%	48m	98.7%

This becomes useful later when TICVAI decides which hardware supplier or model performs better.

AI Insights

Examples:

“Gate Controller model GC-X4 has 2.7× more communication failures than GC-X5.”

“Reader RR-018 has experienced the same network fault four times in 30 days.”

“Preventive maintenance for 16 devices is overdue and correlated with an increased failure rate.”

AI Recommended Actions

Schedule Inspection

Open Maintenance Request

Compare Device Models

Review Vendor

Consider Replacement

AI recommendations remain advisory unless an authorized automation policy exists.

Board 4 Shared Configuration Requirements

Authorized administrators shall be able to configure:

- Maintenance types
- fault categories
- fault reason codes
- severity levels
- priority rules
- preventive maintenance frequencies

- usage-based maintenance thresholds
- maintenance checklists
- device-specific maintenance procedures
- technician/team assignment rules
- vendor assignments
- work-order statuses
- maintenance SLA
- escalation rules
- maintenance windows
- required closure fields
- return-to-service checks
- warranty rules
- warranty expiry alerts
- maintenance cost categories
- attachment requirements
- approval requirements

Maintenance Trigger Architecture

I recommend Softlabs support **four ways of initiating device maintenance**:

1 Scheduled

Preventive maintenance plan becomes due.

2 Manual

Venue/technician reports a fault.

3 Automatic

Board 3 monitoring detects a configured condition.

4 Predictive and AI Recommended

TICVAI identifies a deterioration pattern and recommends maintenance.

The flow should therefore support:

Monitoring and Schedule and User and AI

↓

Maintenance Request

↓

Assessment

↓

Work Order

↓

Diagnosis

↓

Repair

↓

Testing

↓

Return to Service

↓

History and Analytics

Relationship With Module 17

This boundary is important because Module 17 separately contains enterprise asset management, preventive/corrective maintenance, work orders, safety, spare parts and reporting.

We should **not build two independent maintenance engines**.

I recommend:

Module 17 = Central Maintenance Engine

while

Device Manager Board 4 = Device-specific maintenance experience/view.

For example:

A technician creates a work order against **Gate Controller GC-04** from Device Manager.

The same work order should automatically exist in the central **Maintenance & Safety Management module**.

There should be:

One work order ID, one maintenance record, one source of truth.

This will prevent a major architecture duplication later.

RBAC and Governance

Permissions should distinguish:

View Maintenance

Create Request

Create Work Order

Assign Technician

Perform Maintenance

Execute Diagnostics

Restart Device

Manage Warranty

Complete Work Order

Approve Return to Service

Reopen Maintenance

View Maintenance Costs

All critical actions must be audited.

Board 4 Acceptance Criteria

Board 4 is accepted when TICVAI can successfully manage:

Plan → Detect/Report → Request → Work Order → Diagnose → Repair → Test → Return to Service → Track History → Analyze Reliability

for connected devices.

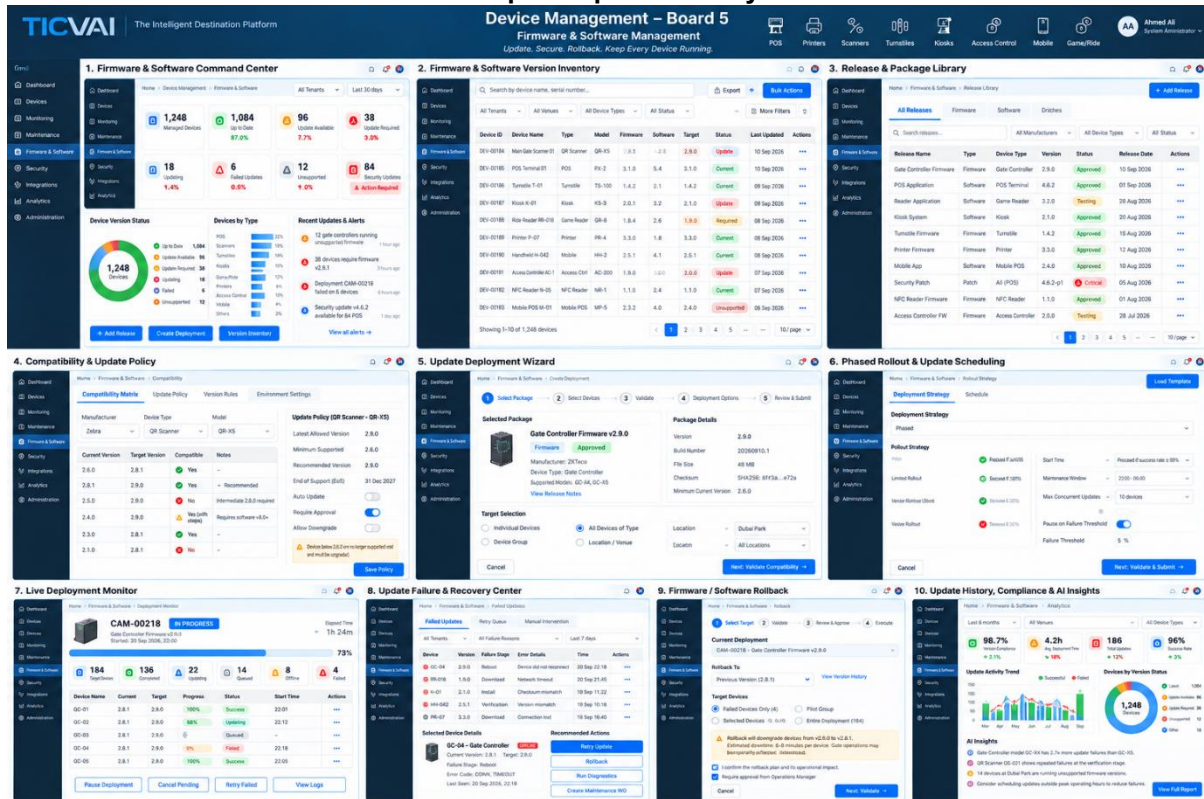
The solution must support preventive and corrective maintenance, work orders, complete maintenance history, warranty tracking and device lifecycle servicing as required by the matrix.

Most importantly, Board 4 must integrate with the future central Maintenance & Safety module rather than creating a second maintenance database or workflow engine.

Board 4 = 10 screens.

Board 5 Firmware and Software Management

Board Objective: Provide TICVAI with a centralized control center to manage the complete firmware and device-software lifecycle across all connected hardware, including **version inventory, approved releases, compatibility, remote deployment, rollout scheduling, deployment monitoring, failures, rollback, and complete update history.**



Board 5 directly covers **16.6 Firmware & Software Management**. The matrix requires firmware management, remote firmware deployment, software-version tracking, software-update management, and update rollback.

The core principle is:

Board 2 manages how a device is configured. **Board 5** manages what firmware/software the device is running and how TICVAI safely upgrades or rolls it back.

This distinction is important because changing a scanner's operating parameters is a **configuration change**, while moving its firmware from v2.8.1 to v2.9.0 is an **update deployment**.

Screen 1 Firmware and Software Command Center

Purpose

Provide a centralized executive and operational view of firmware/software status across the complete device estate.

KPI Cards

Display:

- Total Managed Devices
- Latest Version
- Up to Date
- Update Available
- Update Required
- Update in Progress
- Failed Updates

- Pending/Offline
- Unsupported Versions
- Critical Security Updates

Version Compliance

Example:

1 248 Managed Devices

- 1,084 Up to Date
- 96 Update Available
- 38 Update Required
- 12 Unsupported
- 18 Updating

Breakdown

Analyze by:

Tenant | Venue | Device Type | Manufacturer | Model | Firmware Version | Software Version

Attention Panel

Examples:

- 12 gate controllers are running an unsupported firmware version.
- 38 devices require firmware v2.9.1.
- ⚠ Firmware deployment CAM-00218 failed on 6 devices.
- 🔒 Security update v4.6.2 is available for 84 POS terminals.

Quick Actions

+ Add Release

Create Deployment

Version Inventory

Failed Updates

Update Calendar

Screen 2 Firmware and Software Version Inventory

Purpose

Provide the master inventory showing exactly which firmware and software versions are installed on every device.

Device Version Table

Device	Type	Model	Firmware	Software	Target	Status
GC-04	Gate Controller	GC-X5	2.8.1	4.2	2.9.0	Update
QS-021	QR Scanner	QR-X5	3.4.2	5.1	3.4.2	Current
RR-018	Game Reader	GR-8	1.8.4	2.6	1.9.0	Required

Version Status

Use:

- Current
- Update Available
- Update Required
- Unsupported

☐ Unknown

☒ Exception Approved

Search and Filters

Filter by:

- Tenant
- Venue
- location
- device type
- manufacturer
- model
- firmware
- software
- update status
- support status

Bulk Actions

Select Devices → Update | Schedule | Create Exception | Export

Screen 3 Release and Package Library

Purpose

Maintain the controlled master repository of firmware and software packages approved for deployment through TICVAI.

Release Directory

Release	Type	Device	Version	Status	Released
GC Firmware 2.9.0	Firmware	GC-X5	2.9.0	Approved	10 Sep
Reader App 3.2	Software	GR-8	3.2	Testing	08 Sep
POS Client 4.6.2	Software	POS	4.6.2	Approved	01 Sep

Add Release

Capture:

- Release name
- package type
- manufacturer
- supported device type
- supported models
- version
- build number
- release date
- package/file
- checksum/hash
- digital signature where applicable
- minimum current version
- prerequisites
- release notes

- known issues

Release Classification

Standard

Recommended

Mandatory

Critical

Emergency and Security

Package Status

Draft → **Testing** → **Approved** → **Published** → **Deprecated** → **Withdrawn**

Only approved/published packages may be deployed to production devices.

Screen 4 Compatibility and Update Policy

Purpose

Prevent administrators from deploying incompatible firmware/software to devices.

Compatibility Matrix

Example:

Device Model	Current	Target	Compatible
GC-X5	2.8.1	2.9.0	✓
GC-X4	2.5.0	2.9.0	×
GC-X5	2.6.0	2.9.0	⚠ Intermediate required

Dependency Rules

TICVAI should support rules such as:

Firmware 2.9.0 requires Device Software $\geq$ 4.2.

or:

Device must first upgrade to firmware 2.8.0 before installing 2.9.0.

Update Policy

Configure:

Latest Allowed Version

Minimum Supported Version

Recommended Version

Mandatory Version

End of Support Date

Environment Rules

Allow different policies for:

Development | **Test** | **UAT** | **Production**

This allows TICVAI to test a firmware release before deploying it to live venue devices.

Screen 5 Update Deployment Wizard

Purpose

Create a controlled firmware/software deployment to one or many devices.

Step 1 Select Package

Example:

Gate Controller Firmware v2 9 0**Step 2 Select Target**

Target by:

- Individual device
- selected devices
- device group
- device type
- manufacturer/model
- location
- venue
- tenant

Step 3 Compatibility Validation

Before deployment, TICVAI checks:

- ✓ Correct hardware model
- ✓ Supported current version
- ✓ Required configuration
- ✓ Dependencies satisfied
- ✓ Device reachable
- ✓ Sufficient storage
- ✓ Sufficient battery/power where relevant
- ✓ Security authorization
- ✓ Package integrity

Step 4 Deployment Strategy

Select:

Immediate

Scheduled

Maintenance Window

Phased Rollout

Step 5 Review and Approval

Display:

184 devices selected

178 ready

4 offline

2 incompatible

The user should be able to exclude incompatible devices before proceeding.

Screen 6 Phased Rollout and Update Scheduling**Purpose**

Prevent one faulty firmware release from affecting an entire venue or client estate.

This is especially important for TICVAI because a failed update across hundreds of turnstiles, POS terminals, or game readers could affect live operations.

Rollout Strategy

Support:

Pilot**Limited Rollout****Venue Rollout****Full Deployment****Example****Phase 1 Pilot**

5 devices



Wait 2 hours

**Phase 2 10**

20 devices



Wait 6 hours

**Phase 3 50**

92 devices

**Phase 4 100**

184 devices

Success Gate

Before continuing automatically:

Success Rate 98**Critical Errors = 0****Device Health = Normal****Failure Rule**

Example:

If more than 5% of devices fail, automatically pause the deployment.

Scheduling

Allow:

- Start date/time
- maintenance window
- blackout periods
- venue operating hours
- maximum simultaneous updates

Screen 7 Live Deployment Monitor**Purpose**

Provide real-time visibility while a firmware/software deployment is executing.

Campaign Header**CAM-00218**

Gate Controller Firmware v2.9.0

Progress

184 Target Devices

136 Completed

22 Updating

14 Queued

8 Offline and Pending

4 Failed

Device Level Status

Device	Current	Target	Progress	Status
GC-01	2.8.1	2.9.0	100%	Success
GC-02	2.8.1	2.9.0	68%	Updating
GC-03	2.8.1	2.9.0	—	Offline
GC-04	2.8.1	2.9.0	42%	Failed

Actions

Pause Campaign

Resume

Cancel Pending

Retry Failed

Rollback

Real Time Events

Show timeline:

Package Sent

→ **Downloaded**

→ **Integrity Verified**

→ **Installing**

→ **Restarting**

→ **Reconnected**

→ **Version Verified**

→ **Health Check Passed**

Screen 8 Update Failure and Recovery Center

Purpose

Provide a dedicated workspace for failed or interrupted updates.

Failure Categories

Examples:

- Device offline
- Download failed
- checksum mismatch
- signature invalid
- insufficient storage
- insufficient power
- installation failed
- reboot failed

- device did not reconnect
- compatibility error
- health check failed

Failure Record

Example:

Device: GC-04

Campaign: CAM-00218

From: v2.8.1

To: v2.9.0

Failure Stage: Reboot

Error: Device did not reconnect

Recommended Actions

Retry Update

Rollback

Run Diagnostics

Open Maintenance Request

Create Incident

Mark for Manual Intervention

Integration With Board 4

If hardware/manual intervention is required:

Create Maintenance Work Order

The device, update campaign, error logs and failure information should automatically attach to the maintenance request.

Screen 9 Firmware and Software Rollback

Purpose

Safely return devices to a previously approved version when an update creates operational problems.

Update rollback is explicitly required in the matrix.

Rollback Header

Rollback CAM 00218

Current: v2.9.0

Target Previous: v2.8.1

Select Scope

Single Device

Failed Devices

Pilot Group

Venue

Entire Campaign

Pre Rollback Validation

Check:

- ✓ Previous version available
- ✓ Previous version approved
- ✓ Hardware compatible
- ✓ Configuration compatible

- ✓ Rollback supported by manufacturer
- ✓ Device reachable

Impact Preview

Example:

184 devices affected

Estimated downtime: 6–8 minutes/device

Gate operations may be temporarily interrupted.

Approval

Critical rollbacks may require:

Technical Approval Operations Approval

Emergency Rollback

Provide authorized:

Emergency Rollback

with mandatory reason and complete audit trail.

Screen 10 Update History Compliance and AI Insights

Purpose

Maintain a complete history of firmware/software changes and identify update-related risks across the device estate.

Update History

Date	Device	Previous	New	Campaign	Result
10 Sep	GC-01	2.8.1	2.9.0	CAM-218	Success
10 Sep	GC-04	2.8.1	2.9.0	CAM-218	Rolled Back

Compliance KPIs

Display:

Latest Version Compliance

Unsupported Devices

Mandatory Updates Pending

Update Success Rate

Rollback Rate

Average Deployment Time

AI Insights

Examples:

“Firmware v2.9.0 has a 6.8% failure rate on GC-X4 compared with 0.7% on GC-X5.”

“Devices at Venue B show significantly slower update completion due to network performance.”

“Game Reader firmware v1.8.4 is associated with a higher communication-failure rate than v1.9.0.”

Recommended Actions

Pause Release

Review Compatibility

Update Remaining Devices

Investigate Venue Network

Consider Rollback

Board 5 Shared Configuration Requirements

Authorized administrators shall configure:

- Firmware/software package types
- Manufacturer/model compatibility
- version naming rules
- release classifications
- approval workflow
- supported/minimum versions
- mandatory versions
- end-of-support dates
- deployment permissions
- maintenance windows
- blackout periods
- phased rollout rules
- pilot group size
- maximum concurrent updates
- success thresholds
- automatic pause thresholds
- retry policies
- offline-device behavior
- package validation
- checksum/signature requirements
- rollback policies
- emergency rollback permissions
- update data retention
- notification/escalation rules

Recommended Update Architecture

I strongly recommend Softlabs **not build firmware logic specifically for one hardware manufacturer**.

TICVAI should use a vendor-independent model:

Manufacturer and Vendor Package

↓

TICVAI Device Adapter

↓

Package and Version Manager

↓

Compatibility Engine

↓

Deployment Orchestrator

↓

Device and Controller

↓

Verification and Health Check

This is especially important for the game/ride readers and access-control hardware we discussed. One Chinese reader manufacturer may use one firmware-update protocol, while HID, ScreenCheck, a turnstile controller, or another hardware vendor may use something completely different.

The **TICVAI workflow remains identical**, while the device adapter handles vendor-specific communication.

Firmware vs Configuration

Softlabs should clearly understand this boundary:

Board 2 — Configuration	Board 5 — Firmware/Software
Scan mode	Firmware binary
LED behavior	Embedded software
Sound	Device application version
Sync interval	Operating software
Endpoint	Driver/runtime where applicable
Offline rules	Version package
Operational parameters	Upgrade/rollback

A firmware update may require a compatible configuration version, but they remain **separate governed objects**.

Automatic Recovery Strategy

For critical operational hardware, I recommend supporting configurable recovery logic:

Update

- **Device Reboots**
- **Reconnect Check**
- **Version Verification**
- **Health Check**
- **Operational Test**

If successful:

Deployment Successful

If unsuccessful:

Retry

If still unsuccessful:

Automatic Pause

- **Rollback / Manual Intervention**

For high-impact hardware such as gate controllers and turnstiles, **automatic rollback should only happen when explicitly enabled by the client's policy**.

RBAC and Governance

Permissions should distinguish:

View Versions

Upload Package

Create Release

Approve Release

Create Deployment

Schedule Deployment

Start Deployment

Pause Deployment

Retry Failed Devices

Rollback

Emergency Rollback

Withdraw Release

All sensitive actions must record:

User | Timestamp | Package | Version | Target Devices | Before Version | After Version | Result | Reason | Approval

Board 5 Acceptance Criteria

Board 5 is accepted when TICVAI can successfully:

Register Package → Validate → Approve → Target Devices → Compatibility Check → Schedule/Deploy → Monitor → Verify → Recover → Rollback → Audit

firmware and software updates across supported connected devices.

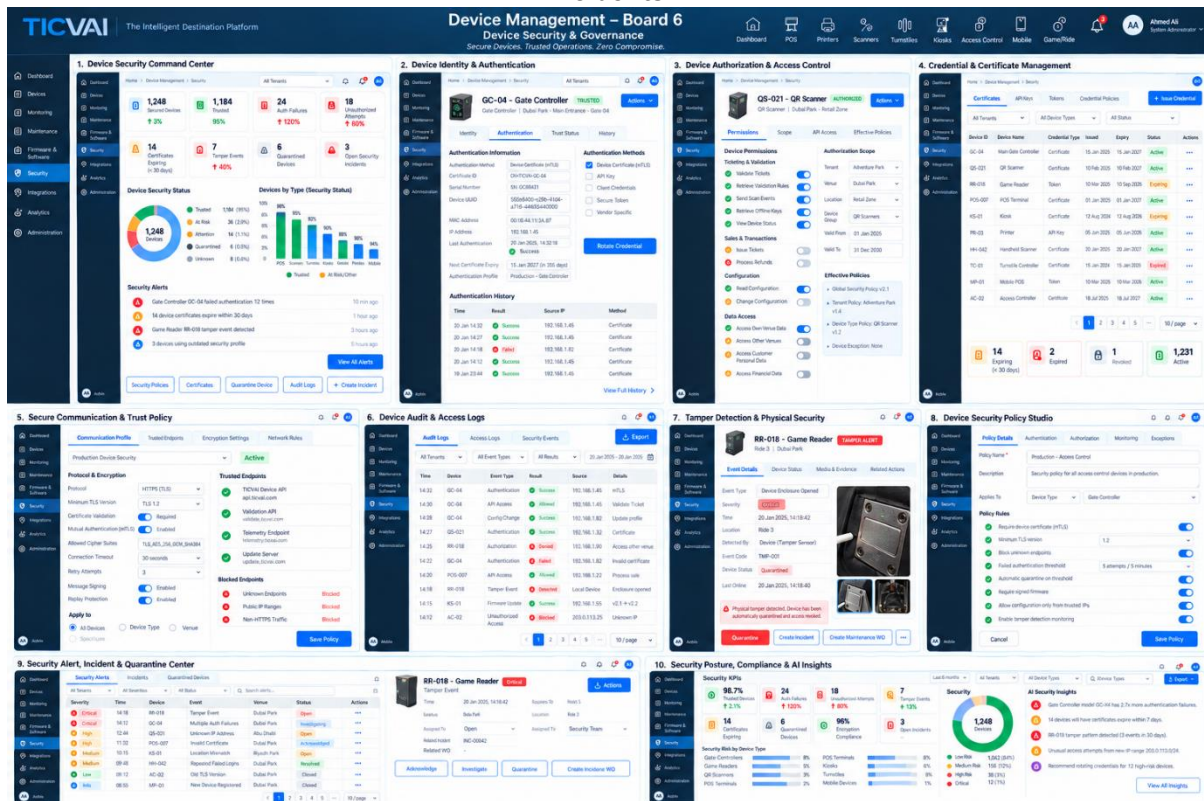
The system must support remote firmware deployment, software-version tracking, update management and rollback as required by the matrix.

It must also support safe phased deployment, device/model compatibility, offline devices, update failures, maintenance integration, complete version history and multi-tenant/venue governance.

Board 5 = 10 screens.

Board 6 Device Security and Governance

Board Objective: Provide TICVAI with a centralized security and governance layer for every connected device, covering **device authentication, authorization, encrypted communication, security credentials, audit logs, access logs, tamper detection, security policies, suspicious activity, and security incidents.**



Board 6 directly covers **16.7 Security**, which requires device authentication, device authorization, secure encrypted communication, device audit logs, device access logs, and tamper detection.

The core principle is:

No device should be trusted simply because it can connect to TICVAI. Every device must have a verified identity, explicit authorization, secure communication, and a traceable security history.

This is particularly important because TICVAI will connect operational hardware such as **POS terminals, scanners, turnstiles, gate controllers, kiosks, NFC readers, game/ride readers and mobile devices.**

Screen 1 Device Security Command Center

Purpose

Provide security, IT, and authorized administrators with a centralized view of the security posture of all TICVAI-connected devices.

KPI Cards

Display:

- Total Secured Devices
- Trusted Devices
- Authentication Failures
- Unauthorized Attempts
- Certificates Expiring
- Security Alerts
- Tamper Events

- High-Risk Devices
- Quarantined Devices
- Open Security Incidents

Overall Security Score

Example:

Device Security Score

96.8 and 100

Break down by:

Authentication | Authorization | Encryption | Credentials | Compliance | Security Events

Device Security Status



Trusted



Attention Required



At Risk



Critical



Quarantined



Unknown

Security by Device Type

Analyze:

POS | Scanner | Turnstile | Gate Controller | Kiosk | Game/Ride Reader | NFC/RFID | Mobile

Critical Attention Panel

Examples:



Gate Controller GC-04 failed authentication 12 times in 10 minutes.

 14 device certificates expire within 30 days.



Game Reader RR-018 reported a physical tamper event.



3 devices are communicating using an outdated security profile.

Quick Actions

Security Alerts

Certificates

Quarantined Devices

Security Policies

Audit Logs

Screen 2 Device Identity and Authentication

Purpose

Manage and verify the unique digital identity of every device connecting to TICVAI.

Device Security Identity

Example:

Device: Main Gate Scanner 01

Device ID: DEV-00184

Security Status: TRUSTED

Identity Information

Display:

- Device ID

- hardware ID
- serial number
- registered MAC where applicable
- device UUID
- certificate identity
- client/application ID
- authentication profile
- last authentication
- trust status

Supported Authentication Methods

TICVAI architecture should support configurable methods such as:

Device Certificate

API Key

Client Credentials

Secure Token

Mutual TLS mTLS

Vendor Specific Secure Authentication

Not every device must use the same authentication mechanism.

Authentication Lifecycle

Registered

→ Credential Issued

→ Identity Verified

→ Authenticated

→ Trusted

Authentication History

Show:

Time	Result	Source	Method
14:32	Success	192.168.1.45	Certificate
14:27	Success	192.168.1.45	Certificate
14:22	Failed	192.168.1.82	Certificate

Screen 3 Device Authorization and Access Control

Purpose

Control exactly what an authenticated device is permitted to do.

Authentication answers:

Who is this device

Authorization answers:

What is this device allowed to do

Device Permissions

Example:

Main Gate Scanner 01

Allowed:

✓ Validate Tickets

- ✓ Retrieve Validation Rules
- ✓ Send Scan Events
- ✓ Retrieve Offline Keys
- ✓ Send Health Telemetry

Not Allowed:

- ✗ Issue Tickets
- ✗ Process Refunds
- ✗ Change Configuration
- ✗ Access Other Venues

Authorization Scope

Permissions may be restricted by:

Tenant

Venue

Location

Device Type

Device

Application and API

Operational Function

Security Principle

Apply:

Least Privilege

A device should receive only the permissions required for its operational role.

Example

A Game/Ride Reader may have:

Read Wallet Balance

Validate Entitlement

Deduct Credits

Send Redemption Result

but it should **not** have access to:

- Customer financial information
- finance reporting
- user administration
- unrelated venues

Screen 4 Device Credential and Certificate Management

Purpose

Centrally manage security credentials used by devices.

Credential Inventory

Device	Credential	Issued	Expiry	Status
GC-04	Certificate	10 Jan	10 Jan 2027	Active
QS-021	Certificate	15 Feb	15 Feb 2027	Active
RR-018	Token	10 Mar	10 Sep 2026	Expiring

Credential Lifecycle**Generate and Issue**→ **Activate**→ **Rotate**→ **Renew**→ **Revoke**→ **Expired****Certificate Actions****Issue****Renew****Rotate****Revoke****Replace****Expiry Alerts**

Configurable alerts:

90 Days**60 Days****30 Days****7 Days****Expired****Automatic Rotation**

Where supported:

Automatically issue and rotate device credentials without manual intervention.

Compromised Credential

Authorized administrator can immediately:

REVOKE CREDENTIAL

The device loses access until a valid credential is issued.

Screen 5 Secure Communication and Trust Policy**Purpose**

Define how devices securely communicate with TICVAI and connected services.

The matrix explicitly requires encrypted device communications.

Communication Profile

Example:

Profile: Production Device Security**Protocol:** HTTPS / TLS**Minimum TLS:** Configurable supported standard**Certificate Validation:** Required**Mutual Authentication:** Enabled where supported**Secure Communication Policies**

Configure:

- Allowed protocols
- encryption requirements
- certificate validation
- trusted endpoints

- approved ports
- API security
- token requirements
- connection timeout
- replay protection where supported
- message signing where required

Endpoint Trust

Example:

Device may communicate with:

- ✓ TICVAI Device API
- ✓ TICVAI Validation API
- ✓ TICVAI Telemetry Endpoint

Blocked:

- ✗ Unknown endpoints

Security Violation

If a device attempts prohibited communication:

Log → Alert → Evaluate Risk → Apply Policy

Screen 6 Device Audit and Access Logs

Purpose

Provide complete traceability of security-sensitive actions involving a device.

The matrix requires both **device audit logs** and **device access logs**.

Audit Log

Capture events such as:

- Device registered
- credential issued
- device authenticated
- authentication failed
- authorization changed
- configuration changed
- firmware updated
- credential revoked
- device quarantined
- tamper event
- device reactivated

Log Fields

Field	Example
Timestamp	10 Sep 2026 14:32
Device	GC-04
Event	Authentication
Result	Failed

Field	Example
Source	192.168.1.45
Actor	Device
Correlation ID	SEC-88291

Access Logs

Track:

Device → Service/API → Action → Result

Example:

GC-04 → Validation API → Validate Ticket → Allowed

Search

Search/filter by:

- Device
- tenant
- venue
- event
- user
- IP
- result
- date/time
- security severity

Export

Authorized security/audit users may export logs according to retention policies.

Screen 7 Tamper Detection and Physical Security Events

Purpose

Manage security events where supported hardware reports physical or logical tampering.

Tamper detection is explicitly required by the matrix.

Tamper Events

Possible events:

- Device enclosure opened
- reader removed
- cable disconnected
- secure element failure
- unauthorized reset
- repeated boot attempts
- unauthorized firmware modification
- device identity mismatch
- location mismatch
- physical tamper sensor triggered

Actual events depend on hardware capability.

Event Example

CRITICAL TAMPER ALERT

Game Reader RR 018

Venue: Dubai Park

Location: Ride 03

Event:

Device Enclosure Opened

Time:

14:18:42

Actions

Acknowledge

Inspect Device

Quarantine

Create Security Incident

Create Maintenance Work Order

Evidence

Store where available:

- device telemetry
- event code
- timestamp
- device identity
- location
- related logs
- technician notes

Screen 8 Device Security Policy Studio**Purpose**

Allow administrators to centrally define security policies for groups of devices.

Policy Example**Production Gate Controller Security****Scope**

All Tenants / Specific Tenant / Venue / Device Type / Device Group

Authentication Policy

- Certificate required
- token expiry
- failed authentication threshold
- credential rotation frequency

Authorization Policy

- Allowed services
- allowed APIs
- operational permissions

Communication Policy

- Encryption required
- trusted endpoints
- allowed protocols

Security Monitoring

Example:

5 Failed Authentication Attempts and 5 Minutes

→ High Alert

10 Attempts and 10 Minutes

→ Critical Alert + Quarantine

Policy Hierarchy

Support:

Global Security Policy

↓

Tenant Policy

↓

Venue Policy

↓

Device Type Policy

↓

Device Exception

Individual exceptions require authorization and should have an expiry date.

Screen 9 Security Alert Incident and Quarantine Center

Purpose

Provide a single operational workspace to respond to security threats involving connected devices.

Security Events

Severity	Device	Event	Venue	Status
Critical	RR-018	Tamper	Dubai Park	Open
Critical	GC-04	Auth Failures	Dubai Park	Investigating
High	QS-021	Unknown IP	Venue B	Open

Security Lifecycle

Detected

→ **Alerted**

→ **Acknowledged**

→ **Investigating**

→ **Contained**

→ **Resolved**

→ **Closed**

Quarantine

Authorized users can:

QUARANTINE DEVICE

Quarantine should:

- Block operational access
- revoke/suspend applicable credentials
- prevent unauthorized API activity
- preserve monitoring connectivity where architecture permits

- retain logs/evidence

Quarantine Status

SECURITY QUARANTINE

Recovery

After investigation:

Verify Identity

Rotate Credentials

Validate Configuration

Validate Firmware

Run Diagnostics

Approve Reconnection

→ **Release from Quarantine**

Integration

If physical repair is required:

Create Maintenance Work Order

If enterprise incident management is configured:

Create Incident

Screen 10 Security Posture Risk and AI Insights

Purpose

Give management and security teams a consolidated view of device security risk and recurring patterns.

Security KPIs

- Trusted Device %
- Authentication Success Rate
- Unauthorized Attempts
- Critical Security Events
- Tamper Events
- Expiring Credentials
- Quarantined Devices
- Policy Compliance
- Mean Time to Resolve Security Events

Risk Analysis

Analyze by:

Tenant

Venue

Device Type

Manufacturer

Model

Security Policy

Device Risk Score

Example:

GC-04 — Risk Score: 78 / 100 — HIGH

Contributing factors:

- 12 authentication failures

- outdated credential
- unexpected IP change
- repeated reconnect behavior

AI Security Insights

AI can surface patterns such as:

“Six gate controllers began authenticating from previously unseen network addresses during the last hour.”

“Game Reader RR-018 has generated three tamper events within 30 days.”

“14 devices will have credentials expire within the next seven days.”

“Device group Kiosk-East has a significantly higher authentication-failure rate than comparable devices.”

AI Recommended Actions

Investigate

Rotate Credentials

Review Security Policy

Quarantine Device

Create Incident

Compare Similar Devices

AI should provide evidence/reasoning behind recommendations and **must not automatically quarantine or revoke credentials unless an explicit automation policy authorizes the action.**

Board 6 Shared Configuration Requirements

Authorized administrators shall be able to configure:

- Authentication methods
- device identity requirements
- credential types
- certificate policies
- credential expiry
- credential rotation
- authorization profiles
- operational permissions
- trusted endpoints
- communication protocols
- encryption requirements
- authentication failure thresholds
- security severity levels
- tamper event types
- security alert rules
- quarantine rules
- security policy hierarchy
- device exceptions
- exception expiry
- audit retention
- access-log retention
- security notifications
- incident escalation

- recovery requirements
- release-from-quarantine approval

Device Trust Architecture

I recommend Softlabs build Board 6 around a **Zero-Trust Device Architecture**.

The conceptual flow should be:

Physical Device

↓

1 Identify

Who are you?

Device certificate / token / secure identity.

↓

2 Authenticate

Can TICVAI verify you?

↓

3 Authorize

What are you allowed to do?

↓

4 Secure Channel

Is communication protected?

↓

5 Monitor

Are you behaving normally?

↓

6 Trust and Restrict and Quarantine

This is much stronger than simply storing an API key on every reader.

Important Architecture for TICVAI Hardware

Because TICVAI will integrate different hardware manufacturers, the security capability of devices will vary.

For example:

Advanced Device

may support:

mTLS + Device Certificate + Secure Element

while a simpler reader may only support:

API Token + Device ID + IP/Network Restrictions

Therefore TICVAI should define a **Security Capability Profile** for each device model.

Example:

Security Capability Reader Model GR X8

Certificate: ✓

mTLS: ✓

Secure Storage: ✓

Tamper Sensor: ✓

Signed Firmware: ✓

versus:

Reader Model GR X2**Certificate:** ✕**mTLS:** ✕**Token:** ✓**Tamper Sensor:** ✕**Signed Firmware:** ✕

TICVAI can then calculate the device's security posture based on what the hardware actually supports rather than expecting every reader to provide identical functionality.

Security Integration With Other Boards

Board 6 should consume information from the earlier boards rather than duplicate it:

Board 1 Registration

Provides verified device identity.

Board 2 Configuration

Provides configuration status.

Board 3 Monitoring

Detects suspicious health/connectivity behavior.

Board 4 Maintenance

Handles physical inspection/repair after security or tamper incidents.

Board 5 Firmware

Confirms firmware/software version and update integrity.

Board 6 Security

Combines these signals to establish:

Device Trust Status

This gives TICVAI a much stronger security model.

RBAC and Governance

Permissions should distinguish:

View Device Security**View Security Logs****Manage Credentials****Rotate Credentials****Revoke Credentials****Configure Authorization****Create Security Policy****Approve Security Policy****Acknowledge Security Alert****Create Incident****Quarantine Device****Release Device from Quarantine****Export Security Logs**

Highly sensitive actions such as:

Credential Revocation

Device Quarantine

Security Policy Override

Release from Quarantine

should optionally require approval depending on tenant governance.

Board 6 Acceptance Criteria

Board 6 is accepted when TICVAI can successfully:

Identify → Authenticate → Authorize → Encrypt → Monitor → Detect → Alert → Investigate → Contain/Quarantine → Recover → Audit

connected devices.

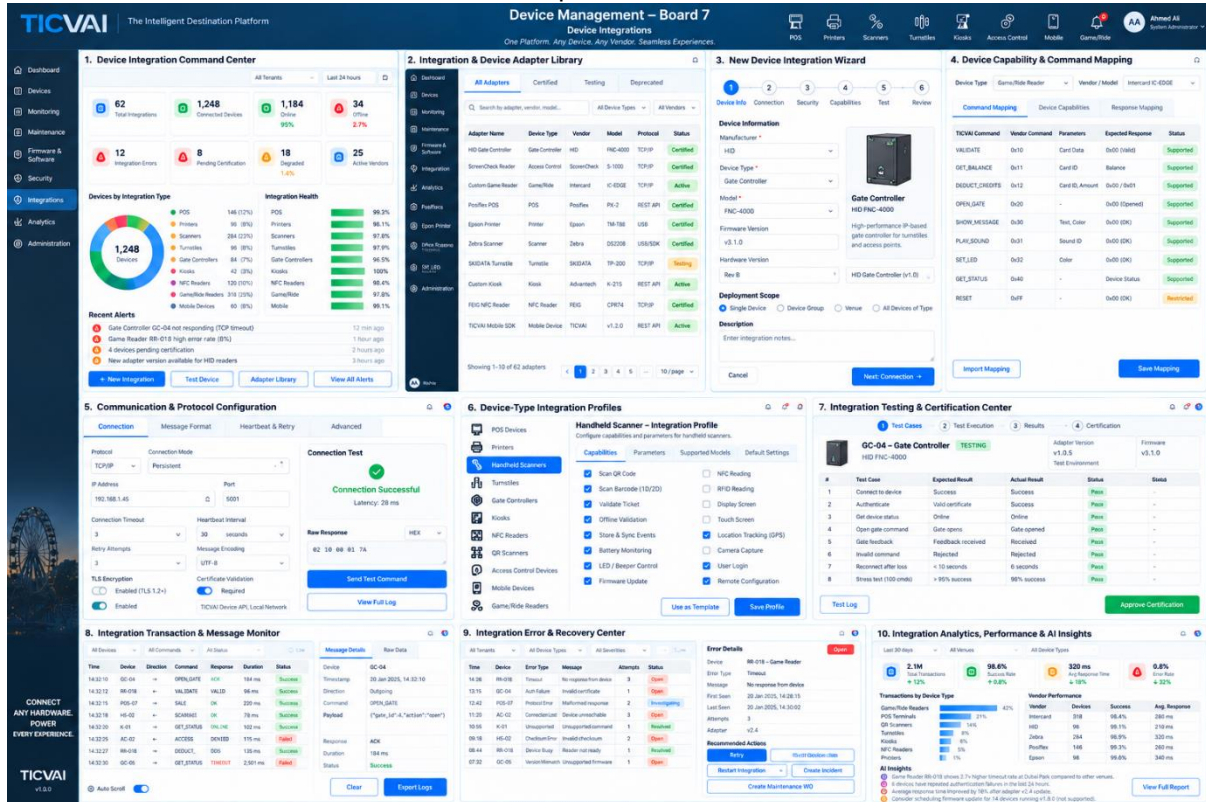
The system must support device authentication, device authorization, encrypted communications, audit logs, access logs, and tamper detection as required by the matrix.

It must also provide configurable device security policies, credential lifecycle management, security-event handling, multi-tenant/venue isolation, device-level risk visibility, and complete security traceability.

Board 6 = 10 screens.

Board 7 Device Integration Hub

Board Objective: Provide TICVAI with a centralized, vendor-independent integration layer for onboarding, configuring, testing, monitoring, and governing all physical devices that communicate with the TICVAI platform.



Board 7 directly covers **16.8 Device Integrations**. The matrix requires integrations for **POS devices, printers, handheld scanners, turnstiles, gate controllers, kiosks, NFC readers, QR scanners, access-control devices, and mobile devices**.

This board is especially important for TICVAI because we should **not build the platform around one hardware manufacturer**.

TICVAI defines the business rules and standard device commands. The Device Integration Layer translates those commands into whatever protocol each manufacturer requires.

This is the same principle we discussed for the Game & Ride module: the reader may come from China or another supplier and provide the screen, LED, NFC/RFID hardware, etc., while **TICVAI owns the backend rules and operational logic**.

Screen 1 Device Integration Command Center

Purpose

Provide one centralized operational dashboard showing all hardware integrations connected to TICVAI.

KPI Cards

Display:

- Total Integrations
- Connected Devices
- Online Integrations
- Degraded Integrations
- Offline Integrations
- Integration Errors
- Pending Certification

- Active Vendors
- API/TCP Connections
- Transactions Today

Integration Health

Example:

Integration	Devices	Online	Errors	Health
QR Scanner	284	279	5	98.2%
Gate Controller	96	94	2	97.9%
Game Reader	318	311	7	97.8%
POS	146	145	1	99.3%
Kiosk	42	42	0	100%

Breakdown

Analyze by:

Device Type | Vendor | Model | Tenant | Venue | Protocol | Integration Status

Attention Panel

Examples:

- 4 Gate Controllers are not responding through TCP/IP.
- Game Reader Adapter v2.4 has an elevated timeout rate.
- ⚠ New Reader Model GR-X10 is awaiting integration certification.

Quick Actions

+ New Integration

Adapter Library

Test Device

Certification Center

Integration Logs

Screen 2 Integration and Device Adapter Library

Purpose

Maintain the centralized catalogue of supported hardware integrations.

Integration Catalogue

Adapter	Device Type	Vendor	Model	Protocol	Status
Gate Controller	Gate Controller	Vendor A	GC-X5	TCP/IP	Certified
Game Reader	Game/Ride	Vendor B	GR-X8	TCP/IP	Certified
QR Scanner	Scanner	Vendor C	QR-500	USB/API	Certified
POS Terminal	POS	Vendor D	PX-2	API	Active
NFC Reader	NFC	Vendor E	NFC-X	SDK	Testing

Adapter Status

- Certified
- Active

- Testing
- ◐ Limited
- ◑ Deprecated
- Unsupported

Adapter Record

Capture:

- Adapter name
- vendor
- manufacturer
- device type
- supported models
- integration method
- protocol
- adapter version
- driver/SDK version
- supported functions
- certification status
- documentation
- technical owner

Important Principle

One adapter can support:

One Model

or

A Family of Compatible Models

This prevents creating unnecessary integrations for every hardware variation.

Screen 3 New Device Integration Wizard

Purpose

Provide a structured workflow for technical administrators to onboard a new hardware integration.

Step 1 Device Information

Capture:

Manufacturer

Vendor

Device Type

Model

Firmware

Hardware Version

Step 2 Integration Method

Select:

- REST API
- WebSocket
- TCP/IP
- MQTT

- SDK
- USB
- Serial
- Bluetooth/BLE
- NFC
- Local service
- Vendor cloud
- Custom protocol

Step 3 Connectivity

Configure:

- Host/IP
- port
- endpoint
- protocol
- timeout
- heartbeat
- retry
- connection mode

Step 4 Security

Select:

- TLS
- mTLS
- certificate
- API key
- token
- device credentials
- network restrictions

Step 5 Capabilities

Define what the hardware supports.

Step 6 Test and Validate

Run integration tests.

Step 7 Certification

Move integration into approved production status.

Screen 4 Device Capability and Command Mapping

Purpose

Define what a particular hardware model can do and map TICVAI's standardized commands to vendor-specific commands.

This is one of the most important screens in Board 7.

TICVAI Standard Command

Example:

OPEN_GATE

Vendor A might require:

CMD=01

Vendor B:

gate.open()

Vendor C:

POST /controller/action/open

TICVAI should not expose these differences to the business modules.

Command Mapping Table

TICVAI Command	Vendor Command	Supported	Response
Validate	CHECK_TICKET	✓	VALID/INVALID
Open Gate	CMD-01	✓	OPENED
Lock Gate	CMD-08	✓	LOCKED
Beep	SOUND-02	✓	OK
Green Light	LED-G	✓	OK
Red Light	LED-R	✓	OK

Capability Profile

Example for a reader:

Display: ✓

Touch Screen: ✓

QR: ✓

NFC: ✓

RFID: ✓

LED: ✓

Sound: ✓

Offline Mode: ✓

BLE: ✕

Configurable Mapping

Administrators/developers can map:

TICVAI Standard Capability

→

Vendor Implementation

This gives TICVAI hardware independence.

Screen 5 Communication and Protocol Configuration

Purpose

Configure how TICVAI communicates with each supported hardware integration.

Connection Profile

Example:

Gate Controller GC X5

Network Configuration

Protocol: TCP/IP

Connection: Persistent

IP: 192.168.1.45

Port: 5001

Timeout: 3 sec

Heartbeat: 30 sec

Retry: 3 attempts

Communication Direction

Support:

TICVAI → Device

Device → TICVAI

or

Bidirectional

Message Configuration

Depending on protocol:

- Header
- payload
- command
- checksum
- terminator
- encoding
- response mapping
- acknowledgement
- error codes

Example

TICVAI sends:

OPEN

Device returns:

ACK-01

TICVAI normalizes this into:

Gate Opened Successfully

Connection Test

Actions:

Test Connection

Send Test Command

View Raw Response

View Normalized Response

Screen 6 Device Type Integration Profiles

Purpose

Provide dedicated integration configuration by device category while maintaining one common architecture.

The matrix explicitly includes ten integration categories.

Supported Profiles

POS Device

Capabilities:

- Transaction initiation

- payment peripheral status
- printer communication
- cash drawer where applicable

Printer

- Print ticket
- print receipt
- paper status
- printer error

Handheld Scanner

- Scan QR/barcode
- validate
- offline validation
- sync events

Turnstile

- Open
- close/lock
- free spin
- direction
- status
- passage confirmation

Gate Controller

- I/O control
- open signal
- feedback
- alarm
- controller status

Kiosk

- ticket sale
- payment
- ticket printing
- peripheral status

NFC Reader

- Read credential
- validate token
- entitlement
- write where supported

QR Scanner

- Decode QR
- validate
- response
- light/sound result

Access Control Device

- validation

- entry
- exit
- anti-passback
- access result

Mobile Device

- device registration
- mobile scanning
- validation
- offline sync

Profile Rule

Only relevant configuration should appear for each device type.

Screen 7 Integration Testing and Certification Center

Purpose

Ensure a hardware model passes TICVAI technical and operational requirements before it is approved for production.

Certification Checklist

Example for a Gate Controller:

- ✓ Device connects
- ✓ Authentication succeeds
- ✓ Heartbeat received
- ✓ Open command successful
- ✓ Gate feedback received
- ✓ Invalid command handled
- ✓ Reconnection successful
- ✓ Offline/recovery tested
- ✓ Security requirements passed
- ✓ Performance requirement passed

Test Cases

Test	Expected	Actual	Result
Connection	Success	Success	PASS
Open Gate	<500ms	220ms	PASS
Feedback	Received	Received	PASS
Reconnect	<10 sec	6 sec	PASS
Invalid Command	Reject	Reject	PASS

Certification Status

Development

→ Testing

→ UAT

→ Certified

→ Production Approved

Certification Record

Store:

- Device model
- firmware tested
- adapter version
- test date
- tester
- test results
- known limitations
- certification expiry/retest requirement

This is especially useful when TICVAI evaluates multiple hardware vendors before choosing the preferred device.

Screen 8 Integration Transaction and Message Monitor

Purpose

Provide technical teams with real-time visibility into communication between TICVAI and connected devices.

Live Message Stream

Example:

Time	Device	Direction	Command	Response	Duration
14:32:10	GC-04	→	OPEN_GATE	ACK	184ms
14:32:12	QR-21	←	QR_SCAN	VALID	96ms
14:32:14	RR-18	←	REDEEM	SUCCESS	122ms

View Modes

Normalized View

Business-friendly:

Ticket Validated → Gate Opened

Technical View

Shows:

- raw request
- raw response
- headers
- payload
- protocol data
- timestamps
- correlation ID

Filters

Filter by:

- Tenant
- venue
- device
- vendor
- protocol

- command
- response
- error
- transaction ID

Security

Sensitive values must be masked.

Example:

API Key: *****

Screen 9 Integration Error and Recovery Center

Purpose

Centralize integration failures and provide controlled recovery actions.

Error Categories

Examples:

- Connection timeout
- device unreachable
- authentication failure
- unsupported command
- malformed response
- protocol error
- invalid checksum
- SDK error
- vendor API failure
- duplicate message
- acknowledgement missing
- version incompatibility

Error Record

Example:

Device: GC-04

Adapter: Gate Controller Adapter v2.4

Error: TCP Timeout

Command: OPEN_GATE

Attempts: 3

Last Response: None

Recovery Actions

Retry

Reconnect

Test Device

Restart Integration

View Logs

Create Incident

Create Maintenance Work Order

Automatic Recovery

Configurable:

Retry 3



Reconnect



Failover where supported



Alert



Escalate

Hardware commands that could cause duplicate operational actions must support idempotency/deduplication where technically possible.

For example, TICVAI must avoid accidentally opening a gate twice or deducting game credits twice because of a retry.

Screen 10 Integration Analytics Performance and AI Insights

Purpose

Provide management and technical teams with visibility into hardware integration quality and vendor/device performance.

KPIs

- Connected Devices
- Integration Availability
- Transactions
- Successful Transactions
- Failed Transactions
- Average Response Time
- Timeout Rate
- Retry Rate
- Error Rate
- Adapter Availability

Vendor Comparison

Example:

Vendor/Model	Devices	Success	Avg Response	Errors
Vendor A GC-X5	96	99.8%	184ms	0.2%
Vendor B GC-X4	64	98.1%	420ms	1.9%
Vendor C GC-X3	48	96.8%	680ms	3.2%

AI Insights

Examples:

- “Gate Controller GC-X5 has 62% faster response time than GC-X4.”
- “Game Reader GR-X8 has a higher timeout rate at Venue B than at other venues.”
- “Adapter v2.4 began showing increased communication failures after firmware v3.1 was introduced.”
- “Six QR scanners are repeatedly reconnecting during peak operating hours.”

Recommended Actions**Review Adapter****Compare Firmware****Inspect Network****Run Certification Test****Create Incident****Review Vendor Performance**

This information can later help TICVAI decide which hardware manufacturers should remain certified/preferred.

Board 7 Shared Configuration Requirements

Authorized administrators shall be able to configure:

- Device integration types
- manufacturers/vendors
- supported device models
- adapter versions
- supported protocols
- IP/port settings
- API endpoints
- SDK/driver references
- heartbeat
- timeout
- retry policies
- command mappings
- response mappings
- error mappings
- device capability profiles
- device-specific functions
- authentication methods
- message formats
- raw/normalized logging
- certification test cases
- certification statuses
- minimum performance requirements
- adapter compatibility
- firmware compatibility
- recovery rules
- error escalation
- integration data retention

Critical TICVAI Architecture Device Abstraction Layer

This is the most important architectural requirement of Board 7.

I recommend Softlabs build:

TICVAI Business Modules

Ticketing / Access / Game & Ride / POS / Kiosk

↓

TICVAI Standard Device API

Examples:

VALIDATE

REDEEM

OPEN_GATE

DISPLAY_MESSAGE

GREEN_LIGHT

RED_LIGHT

BEEP

READ_NFC

PRINT

↓

Device Abstraction and Adapter Layer

Vendor A Adapter

Vendor B Adapter

Vendor C Adapter

↓

Physical Hardware

Gate Controller

Turnstile

Game Reader

QR Scanner

NFC Reader

Printer

POS

Kiosk

This means the Ticketing or Game & Ride module should **never need to know whether the physical reader came from China, HID, ScreenCheck, or another manufacturer.**

It sends the standard TICVAI command.

The adapter translates it.

Example Game and Ride Reader

This architecture fits directly with the Game & Ride approach we established.

The reader itself can provide:

Screen

LED

NFC/RFID

QR where supported

Speaker and Buzzer

Network Hardware

But TICVAI controls:

Price
Credit Deduction
Entitlement
Game and Ride Rules
Session Rules
Access Eligibility
Validation Result
Customer Balance
Usage History

The flow becomes:

Customer taps credential

↓

Reader captures credential

↓

TICVAI validates customer and entitlement

↓

Game and Ride rules execute

↓

TICVAI returns standardized result

↓

Adapter converts result into hardware commands:

VALID

GREEN LIGHT + BEEP + DISPLAY "GO"

or

INVALID

RED LIGHT + ERROR SOUND + DISPLAY REASON

This separation is essential to keep TICVAI's business logic independent from the reader manufacturer.

Example Turnstile and Gate Controller

Similarly:

Access Control Module

decides:

Ticket is valid.

Then sends:

OPEN_GATE

The Device Integration Layer determines how to execute that on the specific controller.

Vendor A:

TCP 192.168.1.45:5001 → CMD01

Vendor B:

POST /gate/open

Vendor C:

SDK.OpenGate(04)

To the Access Control module, they are all simply:

OPEN_GATE

Integration With Boards 1 6

Board 7 should reuse all previous Device Manager capabilities:

Board 1 Device Identity and Registration

↓

Board 2 Device Configuration

↓

Board 3 Monitoring

↓

Board 4 Maintenance

↓

Board 5 Firmware

↓

Board 6 Security

↓

Board 7 Hardware Communication and Integration

Therefore, we should **not** recreate device registration, monitoring, maintenance, firmware, or security inside Board 7.

Board 7 owns the **integration adapter and communication layer**.

RBAC and Governance

Permissions should distinguish:

View Integrations

Create Integration

Edit Adapter

Configure Protocol

Edit Command Mapping

Run Test

View Raw Messages

Certify Device

Approve Production Integration

Restart Integration

Manage Recovery Rules

Deprecate Adapter

Raw protocol/message access should be restricted to technical roles because it may contain sensitive operational information.

Board 7 Acceptance Criteria

Board 7 is accepted when TICVAI can successfully:

Connect → Authenticate → Map Capabilities → Translate Commands → Communicate → Test → Certify → Monitor Transactions → Recover from Errors → Analyze

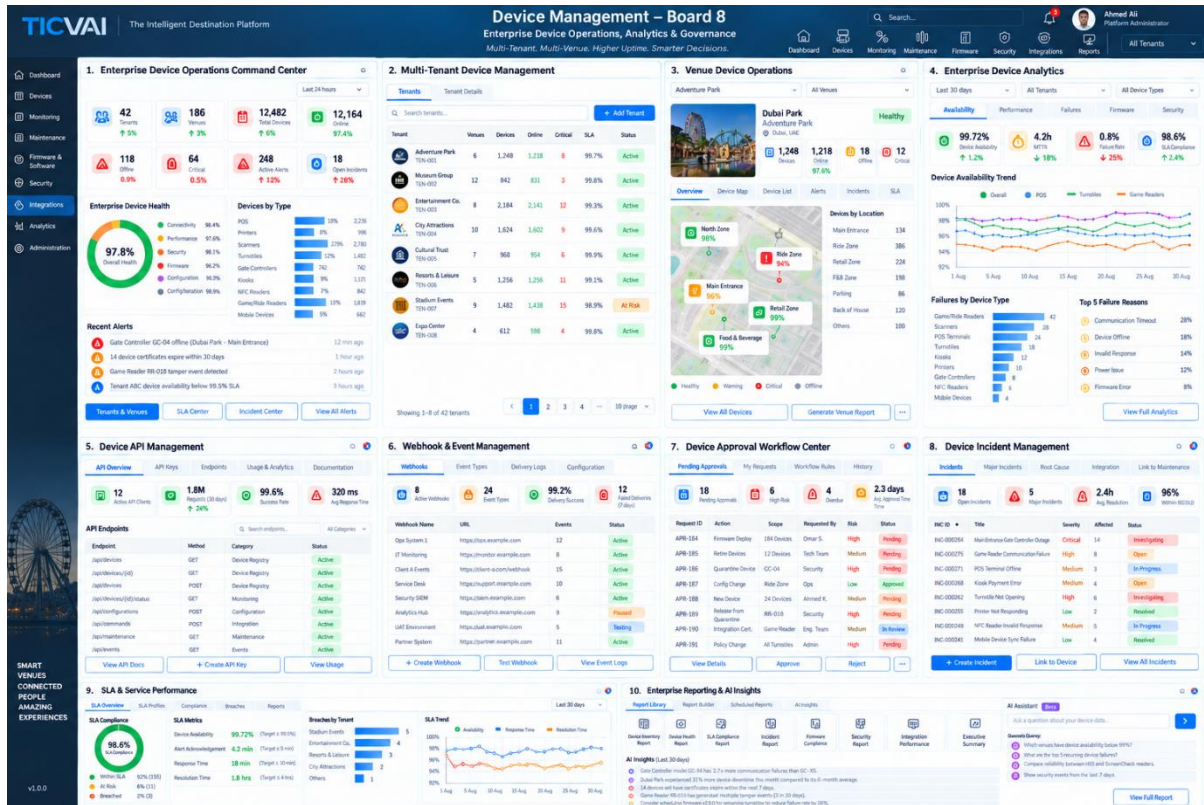
different hardware vendors through a standardized device integration architecture.

The solution must support the matrix-required integration categories: **POS devices, printers, handheld scanners, turnstiles, gate controllers, kiosks, NFC readers, QR scanners, access-control devices, and mobile devices.**

Most importantly, a new hardware manufacturer should be capable of being added primarily through a **new adapter/integration implementation**, without redesigning TICVAI's Ticketing, Access Control, Game & Ride, POS, or other business modules.

Board 7 = 10 screens.

Board 8 Enterprise Device Operations Analytics APIs and SLA



Board Objective: Provide TICVAI with the enterprise management layer for operating the complete device estate across **multiple tenants, clients, venues and locations**, including executive dashboards, tenant/venue governance, analytics, APIs, webhooks, approval workflows, incident management, SLA monitoring and enterprise reporting.

Board 8 directly covers **16.9 Enterprise Features**, which requires multi-tenant device management, venue-specific device management, dashboards, analytics, APIs, webhooks, approval workflows, incident management, SLA monitoring and reporting.

This is the board that brings Boards 1–7 together:

Boards 1–7 manage individual device capabilities. Board 8 gives TICVAI and its clients enterprise-level control over the entire device estate.

Screen 1 Enterprise Device Operations Command Center

Purpose

Provide TICVAI management and authorized client administrators with a consolidated view of device operations across all tenants and venues.

Global KPI Cards

Display:

- Total Tenants
- Total Venues
- Total Devices
- Online Devices
- Offline Devices
- Critical Devices
- Active Alerts
- Open Incidents

- SLA Compliance
- Device Availability

Example:

42 Tenants
186 Venues
12 482 Devices
 12,164 Online
 118 Offline
 64 Critical

Estate Health

Provide:

Enterprise Device Health 97 8

Break down by:

Tenant | Venue | Country/Region | Device Type | Vendor | Model

Operational Attention

Examples:

-  Dubai Park has 14 critical devices.
-  Tenant ABC's device availability has fallen below its 99.5% SLA.
-  18 enterprise incidents remain unresolved.
-  124 new devices were activated this week.

Quick Actions

Tenants and Venues

SLA Center

Incident Center

Analytics

Reports

API and Webhooks

Screen 2 Multi Tenant Device Management

Purpose

Provide strict tenant-level management and isolation for TICVAI's SaaS device estate.

Multi-tenant device management is explicitly required by the matrix.

Tenant Directory

Tenant	Venues	Devices	Online	Critical	SLA
Adventure Park	6	1,248	1,218	8	99.7%
Museum Group	12	842	831	3	99.8%
Entertainment Co.	8	2,184	2,141	12	99.3%

Tenant Profile

Display:

- Tenant name
- tenant ID
- device count

- venues
- device types
- operational status
- support level
- SLA profile
- security policy
- configuration policy

Tenant Isolation

Tenant administrators must only access:

Their Tenant

↓

Their Venues

↓

Their Devices

TICVAI Super Admin can access all authorized tenants.

Cross Tenant Operations

Only explicitly authorized TICVAI roles may:

- Search across tenants
- compare tenants
- manage enterprise policies
- run global reports
- execute cross-tenant actions

Data Isolation

Tenant context must be enforced at:

UI + API + Reporting + Audit + Integration levels.

Screen 3 Venue Device Operations

Purpose

Provide venue-level operations teams with a focused command center for all devices deployed within a specific venue.

Venue-specific device management is explicitly required by the matrix.

Venue Header

Adventure Park Dubai

1,248 Devices

Venue KPIs

- Online
- Offline
- Maintenance
- Critical
- Active alerts
- open incidents
- availability
- SLA compliance

Venue Device Map

Show devices by:

Building → Area → Zone → Location

Example:

Main Entrance

- 12 Turnstiles
- 12 Gate Controllers
- 18 QR Readers

Ride Zone

- 186 Game/Ride Readers

Retail Zone

- 24 POS Terminals
- 16 Printers

Venue Heat Map

Where venue mapping is available:

 Healthy

 Warning

 Critical

 Offline

Venue Actions

View Devices

View Alerts

View Maintenance

View Incidents

View SLA

Generate Venue Report

Screen 4 Enterprise Device Analytics**Purpose**

Provide management with cross-device, cross-venue and cross-tenant analytical capabilities.

Analytics Dimensions

Analyze by:

- Tenant
- venue
- location
- device type
- manufacturer
- model
- firmware
- configuration
- integration adapter
- time period

KPI Library

Examples:

- Availability
 - Health Score
 - Failure Rate
 - MTBF
 - MTTR
 - Alert Rate
 - Maintenance Rate
 - Firmware Compliance
 - Configuration Compliance
 - Security Compliance
 - Integration Success Rate
 - Comparative Analytics
- Example:
- Gate Controller Comparison

Model	Devices	Availability	Failures	MTTR
GC-X5	684	99.82%	12	22m
GC-X4	421	98.94%	38	46m
GC-X3	188	97.81%	31	58m

- Drill Down
- Enterprise
- Tenant
 - Venue
 - Device Type
 - Model
 - Individual Device

AI Query

Allow natural-language analytical queries such as:

“Which gate-controller model had the highest failure rate during the last 90 days?”

“Show venues with device availability below 99% this month.”

“Which game readers generate the most communication errors?”

Results must respect tenant permissions.

Screen 5 Device API Management

Purpose

Expose controlled Device Management capabilities to other TICVAI modules and approved external systems.

Device APIs are explicitly required by the matrix.

API Categories

Examples:

- Device Registry API**
- Get device
 - create/register device
 - update device

- retrieve device status

Monitoring API

- Health
- connectivity
- telemetry
- alerts

Configuration API

- Retrieve configuration
- deploy approved configuration

Integration API

- Device commands
- integration status

Maintenance API

- Maintenance status
- create maintenance request

API Client Management

Capture:

- Client/application
- tenant
- environment
- credentials
- scopes
- allowed endpoints
- rate limits
- expiry
- status

API Usage

Display:

Requests

Successful

Failed

Latency

Rate Limit Events

Security

All APIs must respect:

Authentication Authorization Tenant Isolation Audit

Screen 6 Device Webhook and Event Management

Purpose

Allow TICVAI and external systems to subscribe to important device events instead of repeatedly polling the platform.

Webhooks are explicitly required by the matrix.

Supported Events

Examples:

device.registered
 device.activated
 device.offline
 device.online
 device.health_changed
 device.alert_created
 device.maintenance_started
 device.maintenance_completed
 device.firmware_updated
 device.security_event
 device.retired

Create Webhook

Configure:

Webhook Name

Destination URL

Tenant

Event Types

Secret and Signature

Retry Policy

Status

Delivery Monitor

Event	Endpoint	Result	Attempts	Time
device.offline	Client A	200 OK	1	14:32
device.alert	Client B	Timeout	3	14:31

Retry

Configurable:

Retry → Backoff → Retry → Dead Letter / Failed Event

Actions

Test Webhook

Replay Event

Disable

View Payload

Sensitive data must be masked according to policy.

Screen 7 Device Approval Workflow Center

Purpose

Provide governed approvals for high-impact Device Management actions.

Device approval workflows are explicitly required by the matrix.

Approval Types

Examples:

- Device activation
- device retirement
- configuration deployment

- firmware deployment
- emergency rollback
- security-policy change
- credential revocation
- device quarantine
- bulk device changes
- production integration certification

Approval Inbox

Request	Action	Scope	Requested By	Risk	Status
APR-184	Firmware Deploy	184 Devices	IT Admin	High	Pending
APR-185	Retire Devices	12 Devices	Ops	Medium	Pending
APR-186	Quarantine	GC-04	Security	Critical	Pending

Approval Detail

Display:

Requested Change

Before

After

Affected Devices

Operational Impact

Risk

Requester

Reason

Actions

Approve

Reject

Request Changes

Delegate

Multi Level Approval

Support configurable workflows:

Technical Manager

↓

Operations Manager

↓

Final Approval

depending on action and risk.

Screen 8 Device Incident Management

Purpose

Provide enterprise-level management of operational incidents affecting one or many devices.

Incident management is explicitly required by 16.9.

Incident Sources

Incidents may originate from:

Monitoring Alert**Maintenance****Firmware Failure****Security Event****Integration Failure****Manual Report****External API****Incident Header**

Example:

INC-000284

Main Entrance Gate Controller Outage

CRITICAL

Impact

Tenant: Adventure Park

Venue: Dubai Park

Affected Devices: 14

Affected Location: Main Entrance

Operational Impact: Admission lanes reduced

Incident Lifecycle**Detected**

→ Open

→ Assigned

→ Investigating

→ Mitigation

→ Monitoring

→ Resolved

→ Closed

Incident Workspace

Include:

- affected devices
- alerts
- maintenance work orders
- security events
- firmware changes
- integration errors
- timeline
- owner
- SLA
- root cause
- resolution

Major Incident

Support grouping multiple related device alerts into:

One Parent Incident

instead of creating 100 separate incidents.

Screen 9 Device SLA and Service Performance

Purpose

Monitor whether TICVAI device services meet contractual and operational service levels.

SLA monitoring is explicitly required by the matrix.

SLA Profiles

Configure by:

Tenant

Venue

Device Type

Service Level

Example SLA

Critical Gate Controller

Availability:

99.9

Alert Acknowledgement:

5 minutes

Response:

15 minutes

Resolution:

2 hours

SLA Dashboard

Display:

- SLA compliance %
- availability
- response compliance
- resolution compliance
- breaches
- at-risk incidents

Status

☒ **Within SLA**

☐ **At Risk**

☒ **Breached**

SLA Timer

Example:

INC-000284

Resolution SLA:

01:22:14 remaining

Escalation

Example:

75 SLA consumed

→ Warning

90

→ Escalate

100

→ SLA Breach

SLA Exclusions

Support governed exclusions such as:

- Planned maintenance
- approved outage
- client-caused issue
- force majeure where contractually applicable

All exclusions require auditability.

Screen 10 Enterprise Reporting AI Insights and Executive Summary
Purpose

Provide management, clients and authorized stakeholders with comprehensive reporting across the Device Management module.

Device reporting is explicitly required by the matrix.

Executive Summary

Display:

Device Estate Performance**Availability:** 99.72%**Healthy Devices:** 97.8%**SLA Compliance:** 98.6%**Critical Incidents:** 12**MTTR:** 38 minutes**Firmware Compliance:** 96.2%**Security Compliance:** 98.1%**Report Library**

Provide:

Device Inventory Report**Device Health Report****Availability Report****Maintenance Report****Firmware Compliance Report****Security Report****Integration Performance Report****Incident Report****SLA Report****Device Lifecycle Report****Report Builder**

Users can select:

Metrics

-

Dimensions

-

Filters

-

Date Range

•

Grouping**Delivery**

Support:

View**Export****Schedule****Share according to permissions****AI Insights**

Examples:

“Venue B experienced 31% more device downtime this month compared with its six-month average.”

“Gate Controller GC-X4 is responsible for 42% of all gate-controller maintenance incidents while representing only 18% of deployed controllers.”

“Firmware v2.9.0 reduced communication failures by 26% after deployment.”

“Tenant C is projected to breach its device-availability SLA if the current outage continues.”

AI Query

Management can ask:

“Why did device availability decrease this week?”

“Which vendor is giving us the most problems?”

“Compare ScreenCheck and HID reader reliability.”

“Show all venues at risk of SLA breach.”

AI results should provide drill-down to the supporting operational data.

Board 8 Shared Configuration Requirements

Authorized administrators shall be able to configure:

- Tenant hierarchy
- venue hierarchy
- tenant isolation policies
- enterprise device visibility
- cross-tenant permissions
- enterprise dashboard KPIs
- analytical metrics
- API clients
- API scopes
- rate limits
- webhook events
- webhook retry policies
- approval workflows
- approval levels
- approval thresholds
- incident categories
- incident priorities
- incident ownership

- major-incident rules
- SLA profiles
- availability targets
- response targets
- resolution targets
- SLA exclusions
- escalation thresholds
- report templates
- scheduled reports
- data-retention policies

Enterprise Hierarchy

I recommend Softlabs build the Device Manager around a clear hierarchy:

TICVAI Platform

↓

Tenant and Client

↓

Venue

↓

Building and Area

↓

Zone and Location

↓

Device Group

↓

Individual Device

Permissions, configurations, security policies, monitoring rules, SLA and analytics should be capable of inheriting through this hierarchy.

For example:

TICVAI Global Policy

↓

Adventure Park Tenant Policy

↓

Dubai Park Venue Policy

↓

Main Entrance Device Group

↓

GC 04 Device Exception

This keeps the platform scalable for both a client with **20 devices** and an enterprise client with **20,000+ devices**.

Important Architecture Incident Correlation

I strongly recommend including an **incident correlation engine**.

Example:

At 14:00:

28 devices suddenly go offline at Main Entrance.

Without correlation:

28 separate device incidents.

With correlation:

TICVAI recognizes:

Same venue + same location + same time + similar connectivity failure

and recommends:

Possible Common Infrastructure Failure

Create:

1 Major Incident

with:

28 affected devices

This will dramatically improve operations when TICVAI is managing large venues.

SLA Calculation Principle

SLA should not simply calculate whether a device is online.

TICVAI should distinguish:

Technical Availability

Device is connected.

versus

Operational Availability

Device can actually perform its required function.

For example:

A gate controller might be:

● **Online**

but:

● **Unable to open the gate**

Therefore:

Online Operational

Board 8 should calculate SLA using the applicable service definition.

API and Webhook Architecture

The recommended enterprise architecture is:

TICVAI Device Management Core

↓

Event Bus and Device Events

From Boards 1–7

↓

Two consumption methods:

REST and Device APIs

External systems request information or actions.

and

Webhooks and Events

TICVAI automatically pushes events to subscribed systems.

Example:

Gate Controller goes offline

↓

device.offline

↓

Webhook

↓

Client Service Desk / Monitoring Platform

This makes the Device Manager much easier to integrate with enterprise clients.

RBAC and Governance

Permissions should distinguish:

Enterprise View

Tenant Administration

Venue Administration

Cross Tenant Analytics

API Administration

Webhook Administration

Approve Device Actions

Create and Manage Incidents

Manage SLA

Override SLA Exclusion

Create Reports

Schedule Reports

View Executive Analytics

Every action must respect tenant isolation and maintain a complete audit trail.

Board 8 Acceptance Criteria

Board 8 is accepted when TICVAI can successfully:

Govern Tenants → Manage Venues → Analyze Estate → Expose APIs → Publish Events/Webhooks → Approve High-Risk Actions → Manage Incidents → Track SLA → Report → Generate AI Insights

across the complete Device Management ecosystem.

The solution must support all ten enterprise capabilities required by the matrix: **multi-tenant device management, venue-specific management, dashboards, analytics, APIs, webhooks, approval workflows, incident management, SLA monitoring and reporting.**

Device Management Module Final Coverage

With Board 8, the full **Module 16 – Device Management** is now structurally covered:

Board 1 — Device Command Center & Registration

Board 2 — Device Inventory & Configuration

Board 3 — Device Monitoring, Health & Alerts

Board 4 — Device Maintenance & Lifecycle Servicing

Board 5 — Firmware & Software Management

Board 6 — Device Security & Governance

Board 7 — Device Integration Hub

Board 8 — Enterprise Device Operations, Analytics, APIs & SLA

Total: 8 Boards × 10 Screens = 80 Screens.

This completes the Device Management module without intentionally duplicating the deeper Maintenance & Safety functionality that we will handle under **Module 17**.